Digital Signal Transmission

Reconstruction and Channel Coding

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Abstract

This study evaluates signal reconstruction through Gaussian noise and a binary symmetric channel using standard signal-processing methods. DFT analysis selects 16 kHz sampling for the supplied two-second multitone, retaining all four components while reducing samples by two thirds. The baseline combines a 385-tap Hann multiband FIR, 14-bit endpoint quantisation, Hamming(7,4) and periodic Fourier reconstruction. Purpose-seeded channels use separate random streams under a documented project mapping.

Independent four-configuration comparisons give the lowest anchor-point mean NMSE, 0.029339, when filtering and coding are both enabled. Filtering becomes harmful in Gaussian-only 35/40 dB checks. Seven of 92 R4 means miss the planned 5% relative interval half-width target (maximum 42.43%); some high-SNR contrasts remain unresolved.

Changing to 16 bits improves the declared main-domain objective but raises traffic from 392 to 448 kbit/s, with unresolved anchor-point benefit. Under a separate 320 kbit/s cap, Hamming(15,11) meets the fixed anchor-point mean-quality rule at 305.46 kbit/s, with reduced exact-payload reliability. These conclusions are conditional on the tested signal, severity weights and logical-rate model. Reproducible code, saved measurements and independent numerical-method checks support the results; the numerical assumptions and limitations are stated explicitly.

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1 Reference data and channel models

The bundled 48 kHz reference files contain 96000 mono samples. NPY is real `float64`; WAV is signed 16 bit PCM, related exactly by

$$x_{\mathrm{WAV}}[n] = \mathrm{round}(20000 x_{\mathrm{NPY}}[n])\,. \tag{1}$$

Writing $u = x_{\mathrm{NPY}}$, all experiments use unmodified NPY units; WAV supports listening and content/metadata checking. Both hashes are in `metadata/provided_files.csv`. R1 is deterministic. The simulation uses these channel models:

$$y[n] = x[n] + e[n], \qquad e[n] \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \qquad \sigma^2 = \frac{P_x}{10^{S_{\mathrm{dB}}/10}}, \quad P_x = \frac{1}{N}\sum_{n=0}^{N-1} x[n]^2, \tag{2}$$

$$b_r[n] = b_t[n] \oplus E[n], \qquad E[n] \overset{\text{iid}}{\sim} \mathrm{Bernoulli}(p). \tag{3}$$

`degradation.py` draws standard deviation $\sqrt{\sigma^2}$ without subtracting the realised mean or rescaling. Severity ranges and the root seed are explicit project choices. The following convention separates design trials from held-out validation.

1.1 Random streams and independence convention

Root 2470964, NumPy 2.1.3 `SeedSequence(pool_size=4)` and explicit `PCG64/Generator` fix replay. Purpose branches 0, 1 and 2 separate the following draws:

Purpose	**Seed path**	**Role and indices**
Parameter draws	$(0,0)$, $(0,1)$	S_{ID} in R2.1; p_{ID} in R3.3.
Gaussian draws	$(1, s, \varphi, c, r)$	Study s, design/validation phase φ, condition c and repeat r.
BSC masks	$(2, s, \varphi, c, r, \ell)$	The same study, phase, condition and repeat; code/bit layout ℓ.

Explicit `spawn_key` paths identify the same tree as successive spawning without loop-order dependence [13, 15]. Paired methods share contaminated records; distinct repeats use distinct leaves. Replaying a leaf creates no new observation. Full paths, call shapes, versions, states and input/output hashes are recorded in the run ledgers and `metadata/randomness_design.json`; distribution replay depends on version and calls [14]. Derivations are centralised in Appendix B.

1.2 End-to-end system and evaluation boundary

Figure 1 separates sample processing from transmission and recovery. Gaussian noise enters before the FIR; bit flips enter after ECC encoding. The receiver uses the received bits and public parameters. The original reference is retained only for evaluation. Table 22 collects the frozen baseline and evidence.

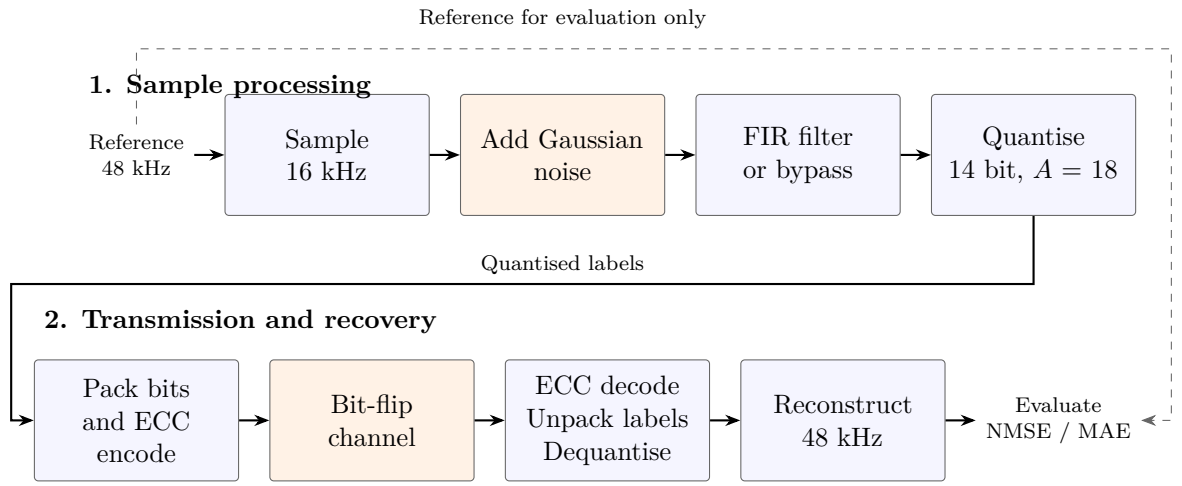


Figure 1: Read each row left to right; the solid return carries quantised labels into the second row. ECC is the fixed Hamming(7,4) code or bypass; FIR/ECC switches give the four R4 configurations. Sampling, quantisation, degradation and reconstruction stay common. The dashed path is measurement only. Formula details are in Appendix B.

2 Stage 1: Signal characterisation and sampling

2.1 Measured spectrum, window and bandwidth

I use the complete unmodified NPY record. Here F_0, L denote reference rate/count and f_s, N transmission rate/count. The reference time and finite-record power conventions give

$$t_n = \frac{n}{F_0}, \quad n = 0, \ldots, L-1, \qquad T = \frac{L}{F_0} = 2\,\mathrm{s}. \tag{4}$$

$$\widehat{P} = \frac{1}{L}\sum_{n=0}^{L-1} x[n]^2 = 0.73, \qquad x_{\mathrm{RMS}} = \sqrt{\widehat{P}} \simeq 0.85440037. \tag{5}$$

The last timestamp is $(L-1)/F_0$, not T. Amplitudes/RMSE retain NPY units; powers/MSE use their squares. The FFT computes the reference DFT with an unscaled forward transform and $1/L$ inverse (Appendix B.1); it does not define a different transform [3]. Native spacing is 0.5 Hz, with no padding. A real FFT displays conjugate pairs once: interior amplitudes are doubled and divided by the window sum, while DC/even Nyquist are not doubled (Eq. (B.4)). Power pairing doubles raw coefficient powers once, not an already doubled amplitude squared.

Choose the window by measured comparison. I retain the rectangular window and cross-check the symmetric Hann of Eq. (B.10), following a controlled window comparison. Hann coherent gain is 0.4999947917; after correction, the four amplitudes differ by at most 3.26×10^{-12} NPY units. Coherence is supported by the phase-aware whole-record fit below and all remaining rectangular bins being below -252.97 dB, not merely by integer indices assigned to detected peaks:

$$\begin{aligned} \phi_j &= \arg X[k_j], \qquad \widehat{x}_m[n] = \sum_j \widehat{A}[k_j]\cos(2\pi f_j n/F_0 + \phi_j), \\ e_{m,\mathrm{RMS}} &= \sqrt{\frac{1}{L}\sum_n (x[n] - \widehat{x}_m[n])^2} = 5.66 \times 10^{-13}. \end{aligned} \tag{6}$$

The measured tones satisfy $f_j T \in \mathbb{Z}$. Hann spreads them into adjacent bins, including 6000.5 Hz; that bin is not a fifth source tone. Rectangular therefore retains the narrow coherent peaks here;

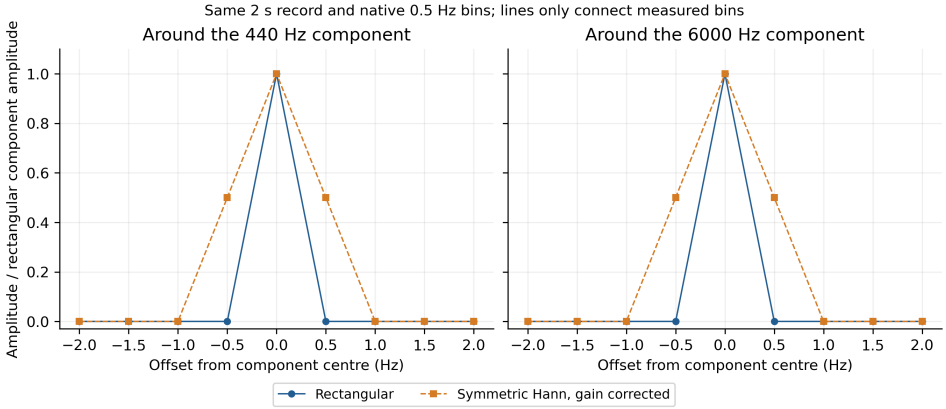


Figure 2: Same-record window comparison on native 0.5 Hz bins. Hann amplitudes are gain-corrected; adjacent skirts are not additional source tones.

Table 1: Measured source components from CSV, in unmodified NPY units. Relative dB uses the largest peak; powers use the paired-coefficient convention.

Bin	Frequency (Hz)	Amplitude	Relative dB	Power (%)
880	440	1.0	0.000	68.4932
2640	1320	0.6	-4.437	24.6575
4400	2200	0.3	-10.458	6.1644
12000	6000	0.1	-20.000	0.6849

this choice does not generalise to noncoherent records. Coherent-gain correction alone does not remove off-bin or overlapping-tone bias [7].

Define and test bandwidth rather than read a plot by eye.

$$A_{\mathrm{ref}} = \max_k \widehat{A}_R[k], \qquad D_w[k] = 20\log_{10}\left(\frac{\widehat{A}_w[k]}{A_{\mathrm{ref}}}\right), \tag{7}$$

$$\mathcal{K}_\theta = \{k : \ k \text{ is a local maximum and } \widehat{A}_w[k] \geq A_{\mathrm{ref}} 10^{\theta/20}\}, \qquad \theta = -60\,\mathrm{dB}, \tag{8}$$

$$f_B = \max_{k\in\mathcal{K}_\theta} f_k = 6000\,\mathrm{Hz}. \tag{9}$$

Interior maxima use left-strict/right-nonstrict comparison, with endpoints tested separately. The operational -60 dB threshold is a project choice; both windows retain the same four centres at $-40, -60, -80$ dB. The weakest tone is about -20 dB. Table 1 gives measured frequency, amplitude and power; Figure 4 combines the full spectrum with the resulting rate decision.

For the alternative cumulative-power definition,

$$B_q = \min\left\{f_k : \frac{\sum_{i=0}^{k} P_i}{\sum_i P_i} \geq q\right\}, \tag{10}$$

$B_{0.99} = 2200$ Hz would discard the valid 6 kHz tone, which contributes 0.685 % of power; $B_{0.999} = 6000$ Hz retains it. I use the 6 kHz component edge to preserve every identified tone. This is finite-record evidence, not strict compact support of its time-truncated DTFT or an unknown analogue source. Tone powers 0.5, 0.18, 0.045, 0.005 sum to 0.73. Ten deterministic checks include DC/Nyquist/odd-length scaling, explicit DFT sums, FFT agreement, inverse, Hann gain and Parseval. Relative Parseval error is 6.08×10^{-16} and maximum same-grid inverse error is 2×10^{-15}; these checks are distinct from low-rate reconstruction.

2.2 Choose 16 kHz with a measured cost comparison

The target is the complete supplied multitone, including its weak highest component. The reference sampling condition and replica geometry are [17]

$$T_s = 1/f_s, \quad x_s[m] = x(mT_s), \quad f_s > 2f_B = 12000\,\text{Hz}, \quad f_N = f_s/2. \tag{11}$$

$$g = f_s/2 - f_B = 2000\,\text{Hz}, \qquad d = f_s - 2f_B = 2g = 4000\,\text{Hz}. \tag{12}$$

The signal's Nyquist rate is 12 kHz; the selected sampler's Nyquist frequency is 8 kHz. The 2 kHz guard is a sampling margin, not a measured filter specification. Following a predict–sample–compare method, I take actual reference values, not newly generated ideal tones:

$$x_M[m] = x_{\text{ref}}[Mm], \quad f_s = F_0/M, \quad N = L/M, \quad t_m = \frac{m}{f_s} = \frac{Mm}{F_0}. \tag{13}$$

$M = 4, 3, 2, 1$ tests 12, 16, 24, 48 kHz. All records span 2 s and have 0.5 Hz native spacing. The implementation is `reference[::factor].copy()`. The phase-aware folding identity is Eq. (B.11); an independent full-complex-FFT comparison of every bin at five measured rates differs by at most 1.12×10^{-16} NPY units.

Table 2: Same 2 s record, common periodic recovery and all 96000 evaluation positions. Raw payload compares equal bits/sample against 16 kHz; 12 kHz is phase-sensitive critical sampling.

f_s (kHz)	M	f_N (kHz)	Samples	Raw ratio	MSE
12	4	6	24,000	0.75	0.005
16	3	8	32,000	1	1.75×10^{-25}
24	2	12	48,000	1.5	1.27×10^{-25}
48	1	24	96,000	3	1.72×10^{-31}

Reconstruction uses only low-rate samples and output length, with the periodic Fourier interpolator of Eq. (B.12). Its L/N scale and split old-Nyquist term are derived in Appendix B.2. Coherent 2 s periodicity supports this boundary model. DC/phased-cosine/Nyquist calibrations and SciPy resampling check it [19]. Every metric uses all 96000 positions (Eq. (B.57)); no trimming, fitted shift/gain or reference-assisted recovery is used. At 16 kHz, MSE/NMSE are $1.75 \times 10^{-25}/2.4 \times 10^{-25}$, peak error 4.39×10^{-12}.

The 12 kHz boundary loses the 6 kHz sine through its measured phase near $-\pi/2$ (Eq. (B.14)): endpoint amplitude 2.05×10^{-13}, MSE 0.005, agreeing with the missing tone power 0.005. Higher 24/48 kHz rates have numerical-residual errors too, but require more payload. Since $48000/M > 12000$ implies $M < 4$, 16 kHz is the lowest sufficient tested integer-decimation choice; it saves one third/two thirds of 24/48 kHz raw bits at fixed B (Eq. (B.40)). The theoretically valid 14.4 kHz alternative would save 10 % relative to 16 kHz with 1.2 kHz guard, but needs noninteger resampling, which I have not measured. I accept 11.11 % extra payload relative to it for direct indexing and a larger guard. This is not a global optimum under later noise.

No appreciable source components lie beyond 6 kHz, so no extra prefilter is inserted in this clean index-selection study. An arbitrary broadband input would require anti-alias filtering. The specified noise is added after downsampling and cannot alias through that earlier operation. Twelve calibration/grid/phase/error checks pass; R1 is deterministic.

2.3 Deliberate undersampling and recovery

The controlled failure bypasses anti-alias filtering at 8 kHz, against the selected 16 kHz path. Removing 6 kHz before sampling would test content loss instead of folding. No stochastic degradation or quantisation is used.

$$x_{16}[m] = x_{\text{ref}}[3m], \qquad x_8[m] = x_{\text{ref}}[6m]. \tag{14}$$

Both last 2 s, with 0.5 Hz bin spacing. The mathematical prediction precedes sample/spectrum checks:

$$f_d = f_0 - q f_s \in [-f_s/2, f_s/2), \qquad f_a = |f_d|, \tag{15}$$

The implementation is $f_d = ((f_0 + f_s/2) \bmod f_s) - f_s/2$. At 8 kHz, 6000 Hz maps to -2000 Hz; its positive 2 kHz cosine phase changes from $-\pi/2$ to $+\pi/2$ (Eq. (B.15)). The 440/1320/2200 Hz tones remain distinct, with no collisions or endpoint ambiguities.

Table 3: Predicted/detected 8 kHz peaks; phase is in units of π. The 6 kHz source folds to positive 2 kHz with reversed cosine phase.

Source (Hz)	Signed (Hz)	Predicted (Hz)	Measured (Hz)	Amplitude	ϕ/π
440	440	440	440	1.0	-0.500
1320	1320	1320	1320	0.6	-0.500
2200	2200	2200	2200	0.3	-0.500
6000	-2000	2000	2000	0.1	0.500

Peaks are detected with the same -60 dB local-maximum rule, rather than declared measured by selecting predicted bins. Normalised complex coefficients and wrapped phases are also checked; all 16 checks pass. Both paths use the same Fourier recovery and all original grid positions. The 8 kHz MSE/RMSE/NMSE/peak error are 0.01, 0.1, 0.0137 and 0.15; the 16 kHz MSE is 1.75×10^{-25}.

Both recoveries fit their own samples to about 10^{-15}, demonstrating that sample fitting does not identify the source. The phase-aware error model predicts 0.1-amplitude tones at 2/6 kHz and MSE 0.01 (Eqs. (B.16)–(B.17)); the full-record error spectrum agrees. This explanation is not information supplied to recovery. Complex coefficients must be added before power when aliases coincide. Later filtering/coding cannot resolve this ambiguity without extra source information.

2.4 Integrated sampling decision

$$\boxed{f_B = 6000\,\text{Hz} \implies f_s > 2f_B = 12000\,\text{Hz} \xrightarrow{f_s = F_0/M,\ M=3} f_s = 16000\,\text{Hz}.} \tag{16}$$

The last arrow is a design choice: the measured 6 kHz edge, tested 16 kHz recovery and 12/8 kHz failure evidence justify it, with the payload and untested-alternative costs stated above.

`stage1/r14/results/decision_summary.csv` joins the five existing rates; six cross-stage checks and seven input hashes verify this reporting join, which introduces no new measurements.

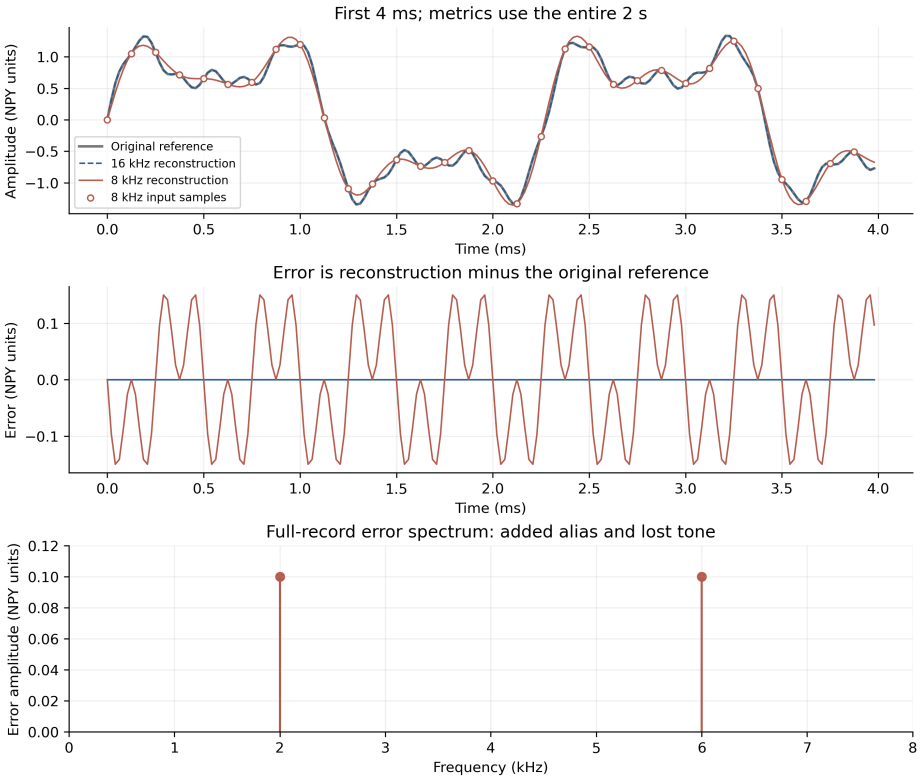


Figure 3: Recovery and error: first 4 ms waveforms above, full 2 s error spectrum below. All 96000 positions enter the reported MSE; sample fitting alone does not identify the source.

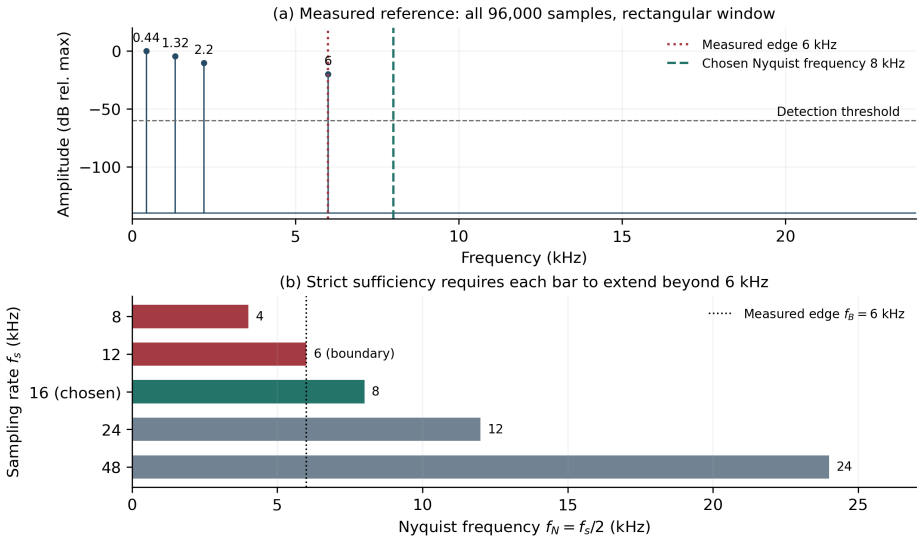


Figure 4: Integrated R1 evidence: full-reference spectrum and candidate Nyquist limits. Peak labels use kHz; the −60 dB detector and −140 dB display floor have separate roles. The selected sampler leaves a 2 kHz guard.

3 Stage 2: Filtering

3.1 Choose a multiband FIR implemented by FFT convolution

I select a multiband FIR because four tones leave wide noise-only gaps. FFT convolution gives the direct-convolution result efficiently; denoising comes from the filter, not the FFT. The implementation follows linear convolution and Fourier analysis; Appendix B.3 derives padding and the centre crop. For the 513-tap prototype, $K \geq N + M - 1 = 32512$; $K = 32768$ is convenient, not mandatory, before cropping to 32000. Padding, identity and zero-filter controls check this implementation.

The input is exactly $u[3n]$ with measured power 0.73. Gaussian degradation follows Eq. (2) after sampling. I declare the anchor-SNR range $[0, 30]$ dB (equal signal/noise power down to one thousandth). Parameter leaf $(0, 0)$ gives

$$U_S = 0.088708911093, \qquad S_0 = 0 + (30 - 0)U_S = 2.661267333 \text{ dB}. \tag{17}$$

Three phase-0 draws at $-10, 0, 10, 20, 30, S_0$ give 18 paired design records, noise paths $(1, 2, 0, c, r)$. All methods share each contaminated array. Personal repeat 0 realises 2.694302 dB; draws are not rescaled. Searching its spectrum within ±10 Hz of declared R1 priors recovers 440/1320/2200/6000 Hz. This is informed design, not blind discovery; the true noise vector never enters construction.

The three-point mean has aligned response $(1+2\cos\omega)/3$. At 6 kHz it is $(1-\sqrt{2})/3 = -0.138071$, attenuating and reversing a valid tone; clean-input MSE is 1.0366×10^{-2}. A broad low-pass also retains much white noise. The prototype instead uses an even four-band ±80 Hz template, a 513-tap IDFT crop and a symmetric Hann kernel taper; a 6600 Hz low-pass is the matched construction comparator. This taper acts on coefficients, not the input record.

Table 4: Method-selection pilot: three paired design draws per level, all 32000 samples. Prototype parameters are provisional; these are design results.

Input SNR (dB)	No filter	Three-point mean	Wide low-pass	Multiband
-10	7.2808×10^{0}	2.4291×10^{0}	5.9946×10^{0}	5.0234×10^{-1}
0	7.2922×10^{-1}	2.5037×10^{-1}	6.0001×10^{-1}	4.9542×10^{-2}
10	7.2617×10^{-2}	3.4625×10^{-2}	5.9907×10^{-2}	5.1871×10^{-3}
20	7.3307×10^{-3}	1.2779×10^{-2}	6.0115×10^{-3}	6.7432×10^{-4}
30	7.3223×10^{-4}	1.0605×10^{-2}	6.0365×10^{-4}	2.5071×10^{-4}

The multiband clean MSE is 2.0139×10^{-4}; fixed 256-sample edges/interior give $1.2550 \times 10^{-2}/5.9998 \times 10^{-7}$, both retained. Its offline prototype has 16 ms causal group delay, not total latency. Direct/FFT outputs agree over 18 inputs within 8.8818×10^{-15}; all 16 checks pass. The reference costs are $O(NM)$ versus $O(K \log K)$ [4], without a measured runtime claim. This pilot selects the family; R2.2 tunes its distortion/noise tradeoff, and R2.3 evaluates unused noise.

3.2 Choose, justify and freeze the filter parameters

I freeze 385 symmetric Hann taps, nominal half-width 70 Hz, delay 192 samples (12 ms) and $K = 32768$, using the measured spectrum and rule below.

Corrupted-spectrum evidence. Spectral analysis of anchor phase-0 repeat 0 uses rectangular, symmetric Hann and Hamming observation windows, distinct from FIR tapers. Equations (B.4)

and (B.9) set amplitude/PSD scales and Parseval. Native PSD uses $K = N$; literal -3.000 dB amplitude widths use 16-fold padding and interpolated crossings:

$$\widehat{S}_{\mathrm{bg}} = \frac{1}{|\mathcal{B}|} \sum_{k \in \mathcal{B}} \widehat{S}_{\mathrm{rect},N}[k], \qquad L_i = 10 \log_{10} \frac{\widehat{S}_{\mathrm{rect},N}[k_i]}{\widehat{S}_{\mathrm{bg}}},$$
$$A_{\mathrm{threshold}} = A_{\mathrm{local\,peak}} 10^{-3/20}, \qquad W_{-3\mathrm{dB}} = f_R - f_L. \tag{18}$$

$\mathcal{B}$ comprises interior positive bins outside ± 20 Hz of the four priors. The mean off-band PSD is 4.9037×10^{-5} squared NPY units/Hz; the 6 kHz peak lies 23.46 dB above it. Padding refines the grid to 0.03125 Hz, not the 0.5 Hz native resolution; power/amplitude ratios use 10/20 log, respectively.

Table 5: Corrupted-spectrum evidence. Analysis-peak widths are Hz, distinct from FIR passbands; peak/background uses native rectangular PSD.

Centre (Hz)	Peak/background (dB)	Rect. width	Hann width	Hamming width
440	43.00	0.4414	0.7185	0.6496
1320	38.71	0.4400	0.7103	0.6432
2200	32.68	0.4432	0.7240	0.6546
6000	23.46	0.4353	0.7207	0.6475

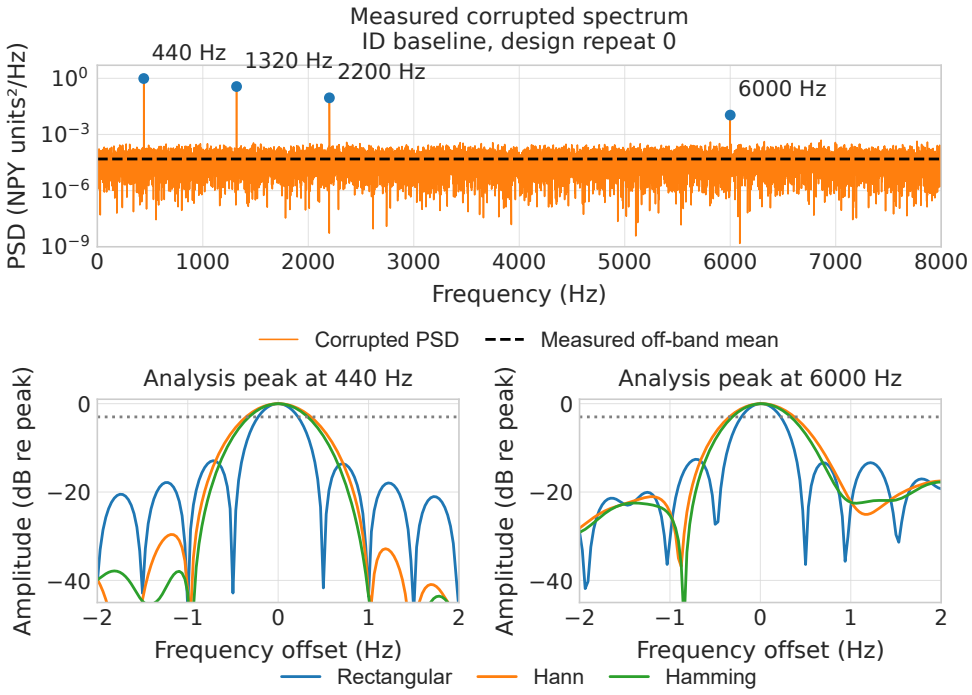


Figure 5: Personal phase-0 spectrum: native PSD/background above; three observation-window peaks and interpolated -3.000 dB widths below.

Sub-hertz observation widths support separated bands; finite-FIR passbands also depend on truncation, gain and boundaries.

Measured search with explicit constraints. The project-designed kernel uses the reference window formulas in Eqs. (B.24) and (B.10). Six lengths 129/257/513/1025/2049/4097, seven half-

widths 2/5/10/20/40/80/160 Hz and three tapers give 126 coarse candidates. Require positive aligned gain within 1% over centre ±2 Hz, checked every 0.5 Hz:

$$G(f) = \sum_{m=0}^{M-1} h[m] e^{-j2\pi f(m-d)/f_s}, \qquad ||G(f_i + \delta)| - 1| \le 0.01. \tag{19}$$

The guard is a design allowance, not measured drift. All candidates share 18 design records; selection uses $\mathcal{C} = \{0, 10, 20, 30, S_0\}$ dB, $R = 3$, and the balanced score

$$J = \frac{1}{|\mathcal{C}|R} \sum_{c \in \mathcal{C}} \sum_{r=0}^{R-1} \frac{\mathrm{MSE}_{\mathrm{out}}(c, r)}{\mathrm{MSE}_{\mathrm{in}}(c, r)}. \tag{20}$$

Each MSE includes 32000 samples; paired normalisation balances SNRs. The −10 dB stress point is excluded. Reference-based scoring supplies no truth to recovery. Refining adjacent bounds around the feasible minimum uses seven odd lengths and 13 widths rounded to 0.5 Hz; with coarse points, 372 candidates include 177 feasible choices. Within 1% of minimum J, prefer shortest length, then lower J, narrower band and registered window order. The winner is also the strict tested minimum and is unchanged at 0/0.5/1% allowances.

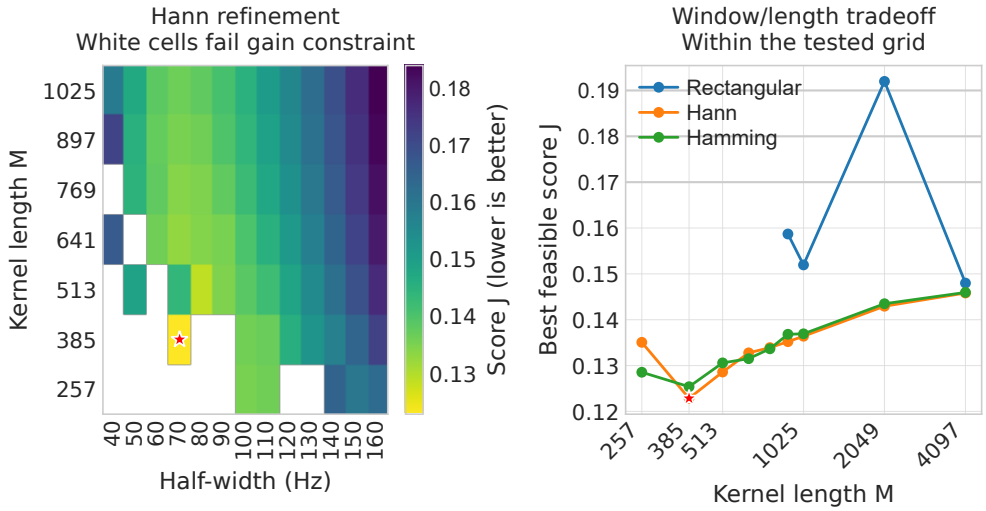


Figure 6: Hann refinement (white: gain-guard failure) and each taper's best feasible tested width at each length. Star: frozen 385-tap choice.

Table 6: Best feasible tested candidate per kernel taper. Scores use the same declared paired design objective, not a universal window ranking.

Kernel window	M	Half-width (Hz)	J	Clean-input MSE
Rectangular	4097	80	0.147986	2.2352×10^{-4}
Hann	385	70	0.122854	2.2951×10^{-4}
Hamming	385	70	0.125388	2.3551×10^{-4}

Result, reason and cost. The four nominal positive bands are $[370, 510]$, $[1250, 1390]$, $[2130, 2270]$ and $[5930, 6070]$ Hz, plus conjugate negative bands. Realised −3 dB full widths are about 103.94 Hz; they are not the 140 Hz template widths. Maximum guard gain deviation is 0.1088%. Relative to the 513-tap prototype, score falls from 0.128593168 to 0.122853660 (4.46%), and group delay from 16 to 12 ms. Both use the same FFT length; runtime gain is

unmeasured. Whole-record buffering is 2 s, so group delay is not total latency. Clean MSE rises to 2.2951×10^{-4}, while finite-record noise gain falls to 0.05322427; at 30 dB design MSE 2.6859×10^{-4} exceeds the prototype's 2.5071×10^{-4}.

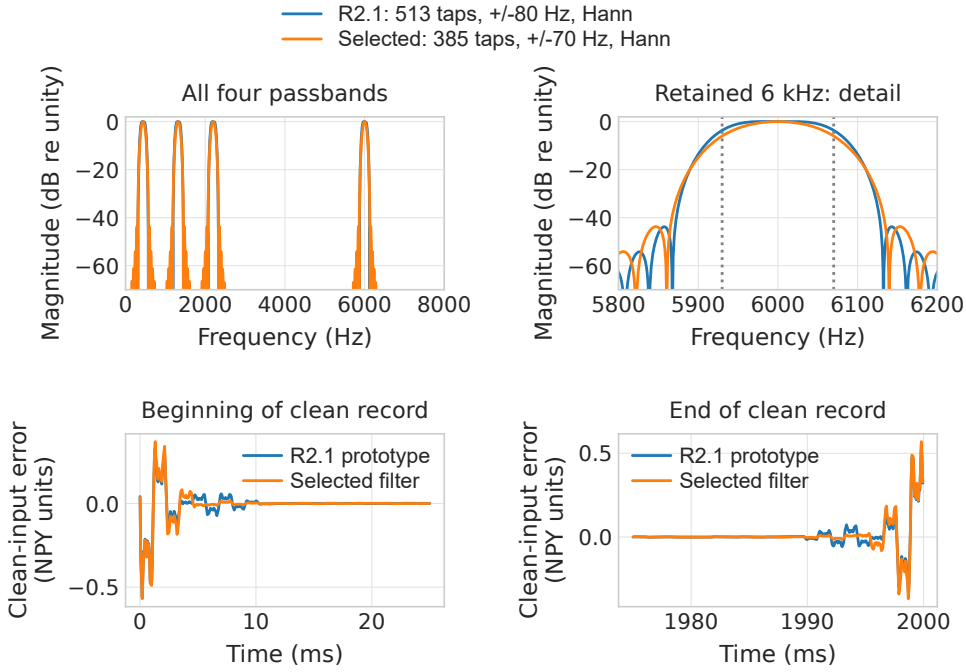


Figure 7: Realised FIR response and retained clean-input endpoint distortion. Template edges and measured −3.000 dB widths differ.

The fixed-filter expectation $D + \sigma^2 G_{\text{finite}}$ predicts a sample-grid crossover near 34.79 dB (Appendix B.3); every trial retains its cross term. Ten checks cover PSD, IDFT, convolution, symmetry and finite-boundary gain. Coefficients/hashes in `stage2/r22/results/filter_freeze.json` precede phase 1.

3.3 Independent recovery assessment and failure boundary

The frozen 385-tap FIR F and bypass receive the identical $y = x + e$. With the unchanged periodic recovery R,

$$z_0 = y, \quad z_F = F(y), \qquad \widehat{x}_0 = R(y), \quad \widehat{x}_F = R(F(y)). \tag{21}$$

All 96000 reconstructed positions enter MSE/NMSE; no fitted gain, shift or trimmed endpoint is allowed. Eq. (B.57) gives units; Eq. (B.59) gives paired reduction and error-SNR gain. The mean gain averages individual logarithmic gains, not the logarithm of mean MSEs. Error SNR includes deterministic distortion and differs from target Gaussian SNR. The signal/error-power approach supports this metric; correlation alone would give one for $\widehat{u} = 2u$ despite MSE P_u.

Before drawing phase-1 noise I fix 50 complete records at each of $-10, 0, S_0, 10, 20, 30, 35, 40$ dB: 400 records, paths $(1, 2, 1, c, r)$. The 35/40 dB diagnostics probe the predicted crossover without changing the declared personal range. Whole records are independent; filtered positions are not extra replicates. Mean intervals use the common approximate t/CLT rule with $R = 50, m = 1$ (Appendix B.7; 20, 21). These pointwise mean intervals are neither simultaneous nor predictions for a new record; count 50 makes no predeclared precision guarantee.

At the anchor point mean MSE falls from 3.9576×10^{-1} to 2.1209×10^{-2} (94.64%); mean paired gain is 12.711 dB, interval $[12.678, 12.743]$. All 50 improve at every tested level through 30 dB,

Table 7: Independent validation, 50 paired records per condition on all 96000 positions. Pointwise approximate 95 % intervals describe mean gain; the dagger marks the personal baseline.

Target S (dB)	Mean MSE, no filter (NPY units^2)	Mean MSE, filter (NPY units^2)	Mean gain [95% CI] (dB)	Wins (/50)
-10	7.2955×10^{0}	3.8900×10^{-1}	12.733 [12.703, 12.762]	50
0	7.3081×10^{-1}	3.8987×10^{-2}	12.730 [12.695, 12.766]	50
2.661†	3.9576×10^{-1}	2.1209×10^{-2}	12.711 [12.678, 12.743]	50
10	7.3101×10^{-2}	4.1467×10^{-3}	12.464 [12.431, 12.496]	50
20	7.3015×10^{-3}	6.1966×10^{-4}	10.714 [10.685, 10.743]	50
30	7.3097×10^{-4}	2.6878×10^{-4}	4.345 [4.329, 4.361]	50
35	2.3069×10^{-4}	2.4202×10^{-4}	-0.208 [-0.219, -0.198]	0
40	7.2974×10^{-5}	2.3326×10^{-4}	-5.047 [-5.055, -5.038]	0

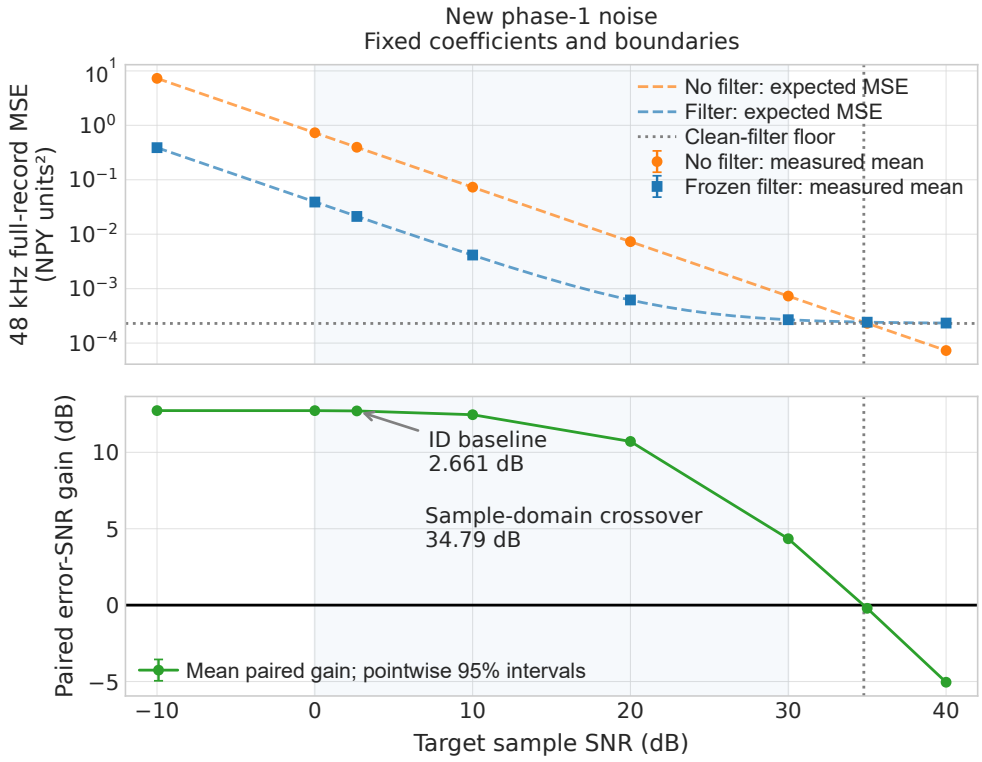


Figure 8: Held-out mean MSE versus prediction (top) and paired gains with pointwise intervals (bottom). Shading: design range; line: prevalidation crossover.

including −10 dB; all 50 worsen at 35 and 40 dB. This is evidence at tested conditions, not every realisation or every intermediate SNR. Fixed FIR distortion persists after the input noise diminishes.

Prediction on the 48 kHz grid includes the old-Nyquist splitting correction (Eq. (B.29)); sample-grid and recovered powers are not silently equated. The no-filter gain is 0.999984375; filtered gain 0.05322427093 and distortion 2.2951×10^{-4} give the plotted $D_{48} + \sigma^2 G_{48}$ prediction. At 40 dB, the first/last 768 positions, only 1.6 % of duration, contribute 97.99% of filtered squared error. Edge shares use aggregated sums, not averaged per-record ratios (Eq. (B.60)); they remain in every primary MSE.

Twelve checks cover inputs, freezing, disjoint seeds, grids and identities. Twenty further audit checks use the separate cosine-series kernel (Eq. (B.31)) and $K = 64000$ real FFT to reproduce 372 scores/6696 design MSEs; the alternative-method replay in Section 7.1 also checks all 400 validation inputs. Records are in `metadata/r2_strict_checks.csv` and `stage2/r23/results/`; these are replays, not extra data or retuning.

4 Stage 3: Quantisation and error control

Phase-0 measurements select the quantiser and code; a separate phase-1 study validates the frozen choices. All results use the supplied signal, the R2 FIR and the receiver in Figure 1.

4.1 Resolution from measured SQNR and fidelity

I select $B = 14$, $A = 18$ and 16384 endpoint levels. The endpoint scale–round–inverse mapping, extended by fixed scaling and saturation, is

$$z_c[n] = \mathrm{clip}(z[n], -A, A), \qquad v[n] = z_c[n]/A, \qquad L = 2^B, \tag{22}$$

$$j[n] = \mathrm{clip}\left(\mathrm{round}\left[\frac{v[n] + 1 - \epsilon}{2}(L-1)\right], 0, L-1\right), \tag{23}$$

$$z_q[n] = A\left(\frac{2j[n]}{L-1} - 1\right), \tag{24}$$

$$q_j = -A + \frac{2Aj}{L-1}, \qquad \Delta = q_{j+1} - q_j = \frac{2A}{L-1}. \tag{25}$$

The implemented spacing is $2A/(2^B - 1)$, whereas standard analysis uses $\Delta_{\mathrm{lec}} = 2A/2^B$. I retain the fixed $\epsilon = 10^{-9}$ and NumPy round-to-even. The shifted granular bound is $|z_q - z_c| \le \Delta/2 + A\epsilon$; zero maps to $-\Delta/2$, so exact zero bias is not assumed. Appendix B.4 derives the codebook, headroom and error model.

The original signal already exceeds $[-1, 1]$. Before drawing noise I fix the same $A = 18$ for filtered and unfiltered inputs over the declared -10 to 40 dB envelope. Six-standard-deviation headroom gives paired-record overload bound 1.2628×10^{-4}; it cannot guarantee no overload for an unbounded Gaussian. Overloads are counted before clipping; none occurred in the 80 branch records. The measured implemented error-power ratio is

$$e_q[n] = z_q[n] - z[n], \tag{26}$$

$$P_z = \frac{1}{N}\sum_{n=0}^{N-1} z[n]^2, \qquad P_{e_q} = \frac{1}{N}\sum_{n=0}^{N-1} e_q[n]^2, \tag{27}$$

$$\mathrm{SQNR}_{\mathrm{meas}} = 10\log_{10}\frac{P_z}{P_{e_q}}. \tag{28}$$

The uniform-error integral, $\Delta^2/12$, and the ideal-model $6.02B + 1.76$ full-range-sine special case are derived in Appendix B.4. They are models, not substitutes for the measured error of this multitone/noisy input occupying part of a wide range.

Measured choice and cost. Five independent design records at each of eight SNRs $\{-10, 0, S_{\mathrm{ID}}, 10, 20, 30, 35, 40\}$, both filter settings and $B = 4, \ldots, 16$ give 1040 measurements. The smallest acceptable width must have no observed overload and satisfy the predeclared full-grid component budget

$$\rho_q(B) = \frac{\mathrm{mean}(q^2)}{\mathrm{mean}(a^2)} \le 0.01 \quad \text{on every design record and both branches.} \tag{29}$$

Here $a = \mathcal{R}(z) - u$ and $q = \mathcal{R}(z_q) - \mathcal{R}(z)$ on all 96000 positions; the even-length Nyquist correction is Eq. (B.29). This 1 % tolerance is a project choice. Thirteen bits reaches 2.2218%, while 14 reaches 0.5586%; the 40 dB no-filter case drives the limit. Budgets 0.1/1/5 % would select 16/14/13 bits, exposing the preference.

Table 8: Fixed-range resolution/cost comparison. Worst component ratio includes both branches and all design records; SQNR is the anchor filtered result.

B (bits)	Personal filtered SQNR (dB, mean of five)	Worst component ratio (%, 80 branch records)	Payload (kbit/s)	Pass 1% budget
8	26.551	2231.9352	128	no
10	38.635	139.9707	160	no
12	50.647	8.8692	192	no
13	56.686	2.2218	208	no
14	62.715	0.5586	224	yes
15	68.732	0.1391	240	yes
16	74.767	0.0347	256	yes

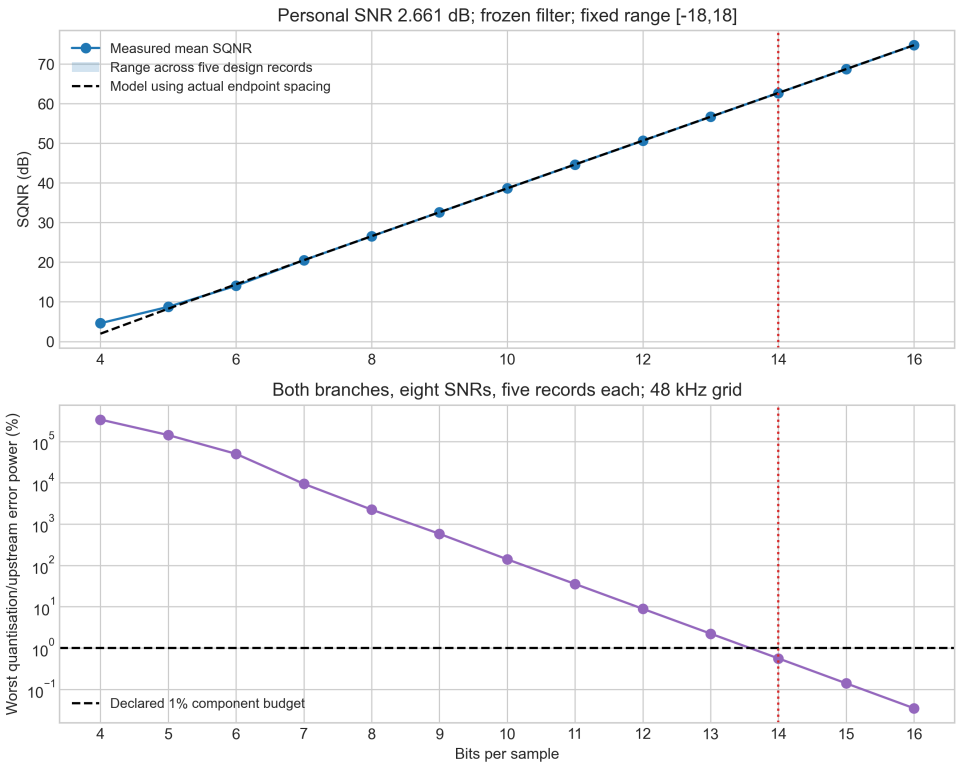


Figure 9: Measured SQNR and the declared component budget. Shading is the five-record design range, not a confidence interval; the low-bit model mismatch is retained.

At the anchor filtered point, measured SQNR is 62.715 dB (range [62.688,62.739]), model mean 62.717 dB and quantisation MSE 4.0259×10^{-7}. Complete-grid MSE changes from 2.1047839×10^{-2} to 2.1048311×10^{-2}. The 1 % component budget permits a 21 % worst-case total-MSE change through cross terms (Eq. (B.61)); the largest observed increase is 0.5840%. A noisier input can have higher SQNR while being less faithful to the reference, motivating the whole-chain metric in R4.

Fourteen bits uses 448000 information bits per record, 12.5 % fewer than 16 bit. Unsigned, sample-major, MSB-first packing is fixed; shared A, B, length and order are simulation metadata, not an assumed protected header. All 131056 labels across the scanned widths round-trip, and independent threshold quantisation/direct convolution/SciPy reconstruction reproduce all 40 noise inputs and 1040 measurements. The quantiser freeze and complete SQNR/error records remain under `stage3/r31/results/`.

4.2 Choose and explain the error-correcting code

I select the taught systematic Hamming(7,4): length 7, four information bits, rate 4/7, $d_{\min} = 3$ and guaranteed correction $t = 1$ [6]. With row messages, encoding is $\boldsymbol{c} = \boldsymbol{m}G \pmod 2$; the receiver computes $\boldsymbol{s} = \boldsymbol{r}H^T \pmod 2$, matches the actual H column, flips that position and extracts the four systematic bits. The reference uses integer matrix multiplication followed by modulo two. The exact G, H, distance proof and worked examples are in Appendix B.5. Its columns are 110, 101, 011, 111, 100, 010, 001; a syndrome integer is not a direct bit index. This code is not SECDED: multiple errors can be miscorrected, and a zero final syndrome does not prove recovery of the transmitted message.

Alternatives, error profile and declared selection. The alternatives are the implemented Hamming(15,11), also $d_{\min} = 3, t = 1$; majority-decoded repetition(5,1), $d_{\min} = 5, t = 2$; and uncoded(1,1). The code constructions and exact decoder expectations are derived in Appendix B.5. A higher Hamming rate does not increase correction capability. Actual iid BSC flips produce binomial block multiplicities, not an artificial one-flip restriction (Figure 10). For the verified decoders, low-p BER begins at $9p^2$ for (7,4), versus $21p^2$ for (15,11).

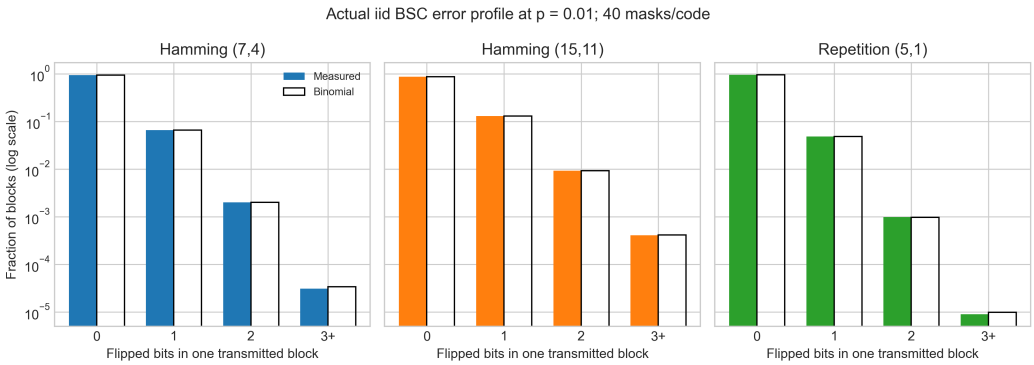


Figure 10: Actual BSC multiplicities versus the binomial model. Both Hamming codes correct one error; repetition corrects two at fivefold traffic cost.

The pilot reuses five R3.1 anchor-SNR inputs, with eight independent masks per positive p/code and one identity control per bank/code: 1140 cases, 1120 positive-p masks. The prespecified probabilities are $0, 10^{-4}, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3$. Five input banks are the independent units for selection; 40 masks are not 40 independent Gaussian inputs. Before BSC generation I fix expansion $M_t/M_b \le 2$ and equal selection weights over $\mathcal{P} = \{0.001, 0.003, 0.01, 0.03\}$:

$$J_{c,g} = \frac{1}{4} \sum_{p \in \mathcal{P}} \frac{\overline{D}_{c,p,g}}{\overline{D}_{\text{uncoded},p,g}}, \qquad J_c = \frac{1}{5} \sum_{g=0}^{4} J_{c,g}. \tag{30}$$

Here $\overline{D}_{c,p,g}$ averages the eight full-grid MSEs in one bank. The normalisation prevents high-p absolute loss alone determining the score. Choose the eligible ECC with smallest $J_c < 1$.

Hamming(7,4) has $J = 0.08344$, approximate pointwise 95 % interval $[0.08133, 0.08555]$, against 0.19204 for (15,11). The paired difference is -0.10860, interval $[-0.11311, -0.10410]$. Repetition

Table 9: Defined alternatives and phase-0 selection score. Feasibility uses the declared twofold expansion budget, not the later rate cap.

Code	n	k	$d_{\min}$	t	Expansion	kbit/s	J
Uncoded	1	1	1	0	1.00000	224	1.00000
Hamming (7,4)	7	4	3	1	1.75000	392	0.08344
Hamming (15,11)	15	11	3	1	1.36366	305.46	0.19204
Repetition (5,1)	5	1	5	2	5.00000	1120	0.01920

improves further ($J = 0.01920$) but needs fivefold expansion. A 1.5/2/5-fold budget would instead select (15,11)/(7,4)/repetition; the twofold budget is a project resource preference, not a physical limit.

Framing, waveform meaning and verification. Equation (B.47) counts real bits, padding and sent bits. H74 uses 784000 bits, no padding, 392 kbit/s; H15 uses 610920 bits, eight padding bits, 305.46 kbit/s. Repetition uses 2240000 bits, 1120 kbit/s. These are logical traffic rates for two seconds of source, not throughput or equal-energy measurements. Padding is removed only after decoding; real-payload BER and padded-block failure have different denominators. Natural-binary bit significance and joint decoded errors mean BER alone cannot rank NMSE (Eq. (B.38)). An independent correction-sphere lookup decoder checks all 32930 received words across the four layouts and replays all 1140 pilot cases. It agrees on payload hashes, errors and full-grid MSE; $p = 0$ recovers the exact prechannel labels. The receiver gets only received bits and public metadata. This frozen, phase-0 choice is subsequently tested on new records in R3.4.

4.3 Choose the error range before the personal draw

The positive main range spans three decades:

$$\mathcal{P}_{\text{main}} = \{10^{-4}, 3\times10^{-4}, 10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}, 10^{-1}\}. \tag{31}$$

I add an identity control $p = 0$, the anchor-seed point, and stress points $0.2, 0.3, 0.5$, giving 12 distinct conditions. Table 10 links this range to the taught BSC and actual code, rather than assuming a physical link's error distribution.

Table 10: Frozen range and stress conditions. Each positive condition exposes 4480000 Hamming blocks; the identity control has 560000. Capacity and finite-code failure are different quantities.

p	Role	$C(p)$	$\Pr(J > 1)$	Expected failed blocks
0	Identity	1.00000	0	0.000
0.0001	Main grid	0.99853	2.0993×10^{-7}	0.940
p_{ID}	Anchor	0.99712	9.3425×10^{-7}	4.185
0.0003	Main grid	0.99606	1.8881×10^{-6}	8.459
0.001	Main grid	0.98859	2.0930×10^{-5}	93.767
0.003	Main grid	0.97054	1.8712×10^{-4}	838.291
0.01	Main grid	0.91921	2.0310×10^{-3}	9099.067
0.03	Main grid	0.80561	1.7093×10^{-2}	76576.793
0.1	Main grid	0.53100	1.4969×10^{-1}	670630.912
0.2	Stress	0.27807	4.2328×10^{-1}	1896308.736
0.3	Stress	0.11871	6.7058×10^{-1}	3004210.944
0.5	Stress	0.00000	9.3750×10^{-1}	4200000.000

At 10^{-4} the exposure predicts 1792 uncoded and 3136 coded raw flips, but only 0.94049 failed Hamming blocks; observing none has probability about 0.3904. Three failures occurred, versus 94 at 10^{-3} (expectation 93.76687). Exact pointwise binomial intervals and theoretical BER accompany these rare counts; zero counts would not prove zero risk (Appendix B.7). Even 4480000 blocks do not provide a precise low-end decoded-BER estimate.

At $p = 0.1$, the actual decoder's block failure probability is 0.1496944: multiple errors are substantial. The rate–capacity crossing is $p = 0.08764841$, while decoded BER loses its advantage at $p = 0.21132487$ (Appendix B.5). They are different boundaries; capacity does not guarantee short-code performance or waveform success [16]. Points 0.2/0.3 straddle the latter. At 0.5, capacity is zero, received bits are independent of the payload and decoded BER is 0.5, providing a severe sanity check. Stress points do not enter the personal mapping.

Reproducible anchor-seed convention. After freezing the range and observation budget, parameter leaf $(0, 1)$ uses NumPy 2.1.3 SeedSequence(pool size 4)/PCG64 with root 2470964:

$$U_p \sim U[0, 1), \qquad \log_{10} p_{\mathrm{ID}} = -4 + 3U_p, \qquad p_{\mathrm{ID}} = 10^{-4+3U_p}. \tag{32}$$

$$U_p = 0.10809119688032798, \qquad p_{\mathrm{ID}} = 0.00021099569319078592. \tag{33}$$

Equal logarithmic intervals have equal weight; the induced density is derived in Appendix B.8. This is a declared project mapping, not an externally specified algorithm or a physical distribution of channels. Each experiment holds p fixed. Code and grid were not changed after seeing the draw. The study marks 570 reused and 320 additional cases explicitly, so its 890 cases/880 positive masks do not double-count observations. Independent replay verifies every payload and error profile; range, parameter state and protocol are frozen before R3.4 validation.

4.4 Measured coding gain under one success rule

Success is acceptable *population mean full-record NMSE*, distinct from individual-waveform or bit-perfect success. Before validation, the phase-0 no-bit-error baseline $d_0 = 0.02883330$ fixes

$$\tau = 2d_0 = 0.05766660, \qquad D_{\max} = P_x\tau = 0.04209662. \tag{34}$$

The factor two is a project preference, not a universal or perceptual limit. Both paths retain the same threshold; the new $p = 0$ mean 0.02906553 does not recalibrate it. Total error retains upstream/channel cross terms (Eq. (B.65)). The 12 R3.3 conditions plus prespecified refinements $0.002, 0.004, 0.005$ are tested with 300 new Gaussian inputs, 8400 positive-p masks and 9000 path records. Whole records are the independent units; Gaussian inputs are shared across probabilities and layouts, so curves are dependent.

$$\text{Gaussian: } (1, 3, 1, 5, r), \qquad \text{BSC: } (2, 3, 1, c, r, \ell), \quad r = 0, \ldots, 299, \quad \ell = 0, 1. \tag{35}$$

The earlier nested-bank variance estimate plans 300 records with half-width target 0.1τ near the two design boundaries; no validation result changes the count. Approximate Bonferroni t/CLT intervals cover the finite 30-mean family (Appendix B.7), not a continuous-p band or individual future records. The common rule and requested gain set are

$$\text{status} = \begin{cases} \text{pass}, & U \le \tau, \\ \text{fail}, & L > \tau, \\ \text{inconclusive}, & \text{otherwise}. \end{cases} \tag{36}$$

$$\mathcal{G}_{\text{tested}} = \{p : \ U_{\text{coded}}(p) \le \tau \text{ and } L_{\text{uncoded}}(p) > \tau\}. \tag{37}$$

Table 11: Independent NMSE means and approximate 30-family intervals, 300 records per path/condition. Gain means coded pass and uncoded fail at the same fixed threshold; decisions use unrounded CSV.

p	Uncoded mean [adjusted CI]	Hamming mean [adjusted CI]	Gain
0	0.0291 [0.0289,0.0292]	0.0291 [0.0289,0.0292]	no
0.0001	0.0873 [0.0825,0.0922]	0.0292 [0.0289,0.0294]	yes
p_{ID}	0.1516 [0.1449,0.1582]	0.0292 [0.0290,0.0294]	yes
0.0003	0.2068 [0.1990,0.2146]	0.0292 [0.0290,0.0295]	yes
0.001	0.6223 [0.6087,0.6360]	0.0333 [0.0321,0.0346]	yes
0.002	1.2219 [1.2007,1.2431]	0.0436 [0.0416,0.0457]	yes
0.003	1.7901 [1.7653,1.8149]	0.0638 [0.0603,0.0673]	no
0.004	2.3823 [2.3534,2.4112]	0.0922 [0.0877,0.0967]	no
0.005	2.9606 [2.9275,2.9936]	0.1196 [0.1143,0.1249]	no
0.01	5.8943 [5.8475,5.9411]	0.3970 [0.3866,0.4074]	no
0.03	17.2616 [17.1866,17.3365]	3.1534 [3.1229,3.1839]	no
0.1	53.2739 [53.1572,53.3906]	27.3053 [27.2175,27.3931]	no
0.2	94.8705 [94.7203,95.0206]	75.2253 [75.0877,75.3630]	no
0.3	124.6956 [124.5513,124.8398]	115.9078 [115.7615,116.0540]	no
0.5	148.9787 [148.8422,149.1151]	149.0071 [148.8766,149.1377]	no

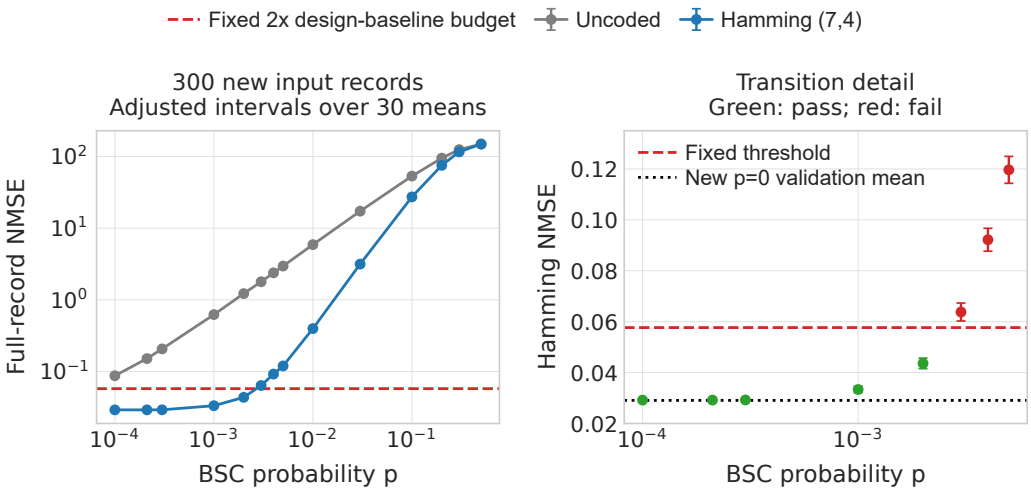


Figure 11: Coded/uncoded mean quality over the full envelope (left) and the coded transition detail (right). Error bars use the frozen 30-mean family; the threshold is fixed before validation.

The primary measured gain is

$$\boxed{\mathcal{G}_{\mathrm{tested}} = \{0.0001,\ p_{\mathrm{ID}},\ 0.0003,\ 0.001,\ 0.002\}}. \tag{38}$$

At the anchor point, coded/uncoded mean NMSE is 0.02917836/0.15156122; the respective intervals lie below and above τ. The largest coded passing test point is 0.002; 0.003 fails. Uncoded $p = 0$ passes, whereas 10^{-4} fails. These are tested brackets, not exact continuous crossings. At higher p both fail quality even where coding still reduces BER. The ancillary mean paired reconstruction-error gain is 7.03379 dB, pointwise 95 % interval $[6.91444, 7.15313]$; it is not an E_b/N_0 gain or the log ratio of two mean MSEs.

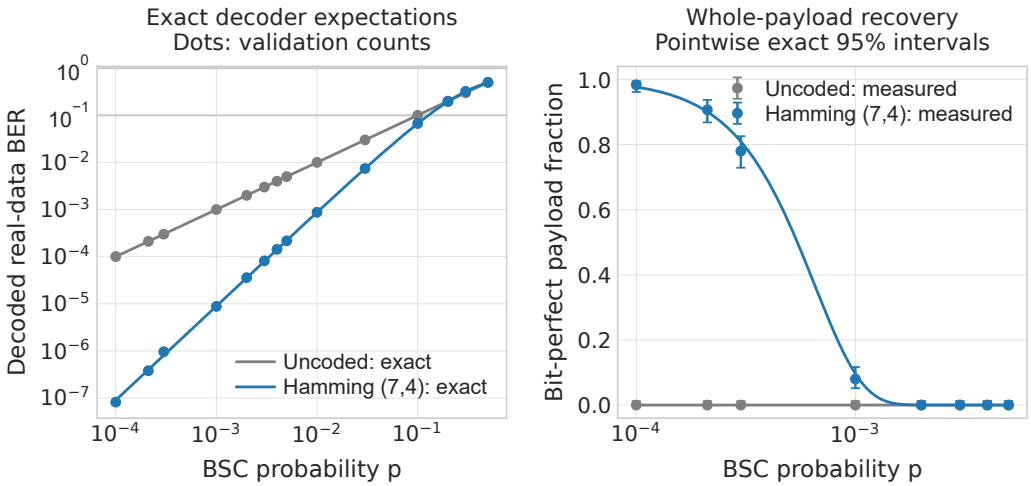


Figure 12: BER versus exact decoder expectations (left); complete 448000-bit payload recovery with pointwise 95% proportion intervals (right). Zero BER counts are omitted from the log axis.

Three meanings of success and the resource cost. At $p = 0.002$ the coded mean passes, 264/300 individual waveforms pass, and 0/300 complete payloads are bit-perfect. At the anchor point all 300 coded waveforms pass but only 272/300 payloads are perfect; both uncoded counts are zero. The coded perfect-payload estimate is 90.67%, pointwise exact 95 % interval $[86.79, 93.71]$%; its theoretical value is 90.07% (Eq. (B.56)). These digital-message probabilities differ from the 30-family waveform mean intervals. The exact BER/block laws describe actual independent BSC flips, not an imposed single-flip experiment.

Prespecified threshold factors 1.5, 2, 4 retain exclusive gain at p_{ID}, 0.0003 and 0.001. Other classifications can change: an interval crossing the threshold remains inconclusive, not silently a pass. The coded stream sends 784000 bits versus 448000 uncoded: 75 % overhead, 392 versus 224 kbit/s, within the twofold budget. Equal p is not equal traffic or energy. Alternative-method replay (Section 7.1) verifies all 9000 paths/decisions within MSE 4.27×10^{-14}; no overload was observed. This establishes a conditional result for the supplied record, fixed anchor SNR, framing and budget, rather than universal code optimality.

5 Stage 4: End-to-end measurement and evaluation

5.1 Why whole-record NMSE, with MAE as an alternative

I retain $D = \mathrm{mean}[(\widehat{x} - u)^2]$ and $d = D/P_u$, $P_u = 0.73$, over all 96000 original positions. The definition measures full-record squared reconstruction error relative to the original reference power. Appendix B.6 derives the common definitions, bias identity, MAE/RMSE relation and dB conventions. NMSE is dimensionless, lower is better and it can exceed one; amplitude gain, timing and endpoints are not fitted or trimmed. Centred variance alone would hide fixed bias.

MAE is a genuinely different loss; RMSE is a monotone transformation of per-record MSE. Figure 13 shows why the choice matters: diffuse error has $(D, \mathrm{MAE}) = (0.01, 0.1)$, sparse large errors $(0.04, 0.02)$. MAE prefers the latter, NMSE the former. Natural-binary MSB errors are 2^{13} times the amplitude of LSB errors at 14 bit (2^{26} in squared error), supporting a loss sensitive to large corruption. These local ratios do not determine total NMSE.

MAE remains a diagnostic; neither metric is claimed as perceptual quality [23]. Correlation alone is insufficient: $\widehat{x} = 2u$ has correlation one but NMSE one. Likewise exact bit delivery

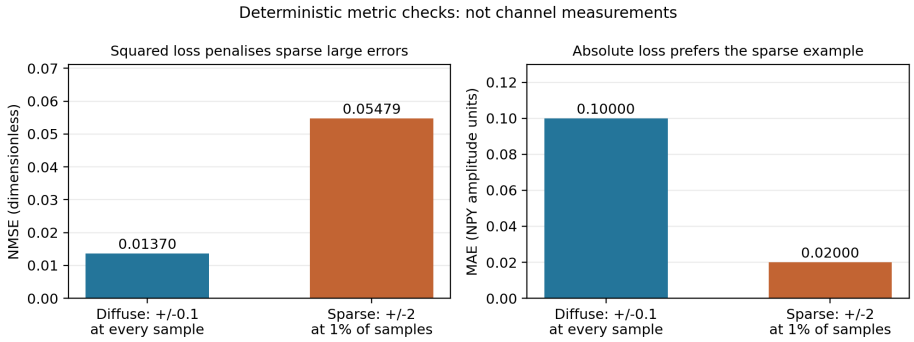


Figure 13: Declared full-grid metric counterexamples. NMSE favours diffuse small error, MAE the sparse large error. Axes retain their different units; this is not a stochastic channel experiment.

retains upstream waveform error. For repeated records I distinguish $-10\log_{10}\bar{d}$ from the mean of individual dB ratios. Seven semantic checks/221 independent assertions and the saved R3 examples verify these interpretations.

5.2 Four configurations and what their contrasts isolate

Filter switch f and code switch c vary while sampling, Gaussian/BSC models, quantiser, framing and receiver remain common. Here x contains 32000 transmission samples; $\mathcal{R}$ returns 96000 reference-grid positions and d_{fc} denotes their full-record NMSE:

$$\begin{aligned} y &= x + \eta, \quad z_f = F_f(y), \quad b_f = \mathrm{pack}(\mathcal{Q}(z_f)), \\ t_{fc} &= E_c(b_f), \quad r_{fc} = t_{fc} \oplus \varepsilon_c, \\ \widehat{x}_{fc} &= \mathcal{R}\Big[\mathcal{Q}^{-1}(\mathrm{unpack}(D_c(r_{fc})))\Big]. \end{aligned} \tag{39}$$

Here $F_0 = E_0 = D_0 = I$; F_1 is the frozen FIR and E_1, D_1 the taught Hamming(7,4). Neither bypasses the protective modules, not the degradations. Code-only quantises noisy unfiltered samples; filter-only sends unprotected bits through the BSC. The receiver has only received bits and public codebook/framing/output-length metadata, never the reference or true masks.

Table 12: Four actual paths carrying 448000 real information bits. Gaussian noise, quantisation, BSC and reconstruction remain enabled in every configuration.

ID	Configuration	f	c	Sent bits / logical kbit/s
00	neither	0	0	448000 / 224
10	filter-only	1	0	448000 / 224
01	code-only	0	1	784000 / 392
11	both	1	1	784000 / 392

One Gaussian vector is shared within each (S, r) across all four paths and p values. Same-code filter pairs replay the same BSC leaf and shape; uncoded/coded layouts have independent masks and unequal lengths.

$$\begin{aligned} \text{Gaussian} &: (1, 4, a, \text{SNR ID}, r), \\ \text{BSC} &: (2, 4, a, \text{joint ID}, r, \text{layout ID}). \end{aligned} \tag{40}$$

At $p = 0$ no mask is drawn. The actual decoder's translation identity (Eq. (B.54)) gives equal XOR error patterns/BER within 00/10 and 01/11, while their waveform NMSE can differ through

bit values, upstream errors and amplitude-error signs. The conditional reductions and interaction are

$$\begin{aligned} g_{F|c=0} &= d_{00} - d_{10}, & g_{F|c=1} &= d_{01} - d_{11}, \\ g_{C|f=0} &= d_{00} - d_{01}, & g_{C|f=1} &= d_{10} - d_{11}, \\ I &= d_{00} - d_{10} - d_{01} + d_{11}. \end{aligned} \tag{41}$$

Positive reductions improve quality; negative values are retained. Comparisons at fixed c isolate enabling the filter; those at fixed f isolate enabling ECC including its extra transmitted bits. Comparing only 00 with 11 confounds these effects. Interaction is not a responsibility percentage, and these switches do not separately identify changing the sampling rate or bit depth. Rates remain 224/392 kbit/s, not equal energy.

Twenty new pilot bundles over 22 unique conditions produce 1760 paths. All are independently replayed; mask invariants and clean/identity controls pass. Pilot variance alone plans 1400 validation bundles for 88 condition means plus four main-objective means, targeting 5 % relative half-width. The smallest passing multiple of 50 is frozen before phase 1 (Eq. (B.80)); a pilot prediction is not a guarantee of achieved precision. No pilot observation enters the final means.

5.3 Central degradation curves and conditional contributions

Figure 14 is the central end-to-end result: all four actual configurations on two severity slices, holding $p = p_{\mathrm{ID}}$ while varying SNR and $S = S_{\mathrm{ID}}$ while varying p. Eight SNR and 15 BSC conditions share one anchor point, hence 22 unique conditions; this is not a full two-dimensional response surface. The 1400 held-out bundles use disjoint phase-1 leaves:

$$\text{Gaussian} : (1, 4, 1, s_G, r), \qquad \text{BSC} : (2, 4, 1, s, r, c), \quad r = 0, \ldots, 1399. \tag{42}$$

There are 123200 paths,11200 unique Gaussian vectors and 58800 positive-p masks. Reused inputs do not multiply independent exposure. At $p = 0$ Gaussian noise remains; 00=01 and 10=11 exactly. Four no-Gaussian/$p = 0$ controls run once outside the random means.

Approximate Bonferroni t/CLT intervals use 1400 complete bundles and the prespecified 92-statistic family (Eq. (B.68)); they are mean intervals, not per-record spread or exact finite-sample guarantees. The 12-condition main objective first averages each bundle's 12 equally weighted, unique main conditions, then estimates uncertainty from 1400 bundle scores (Eq. (B.77)). Diagnostic/refinement points have zero main weight; the personal overlap is counted once. The weights are a design objective, not observed channel probabilities.

Table 13: Held-out anchor-point results. Only NMSE intervals belong to the 92-mean family; MAE and real-information BER are descriptive.

Config.	Mean NMSE	Adjusted mean interval	Mean MAE	Data BER
00	0.664309	[0.660879, 0.667739]	0.513143	2.1117×10^{-4}
10	0.151933	[0.148476, 0.155391]	0.131631	2.1117×10^{-4}
01	0.541989	[0.541564, 0.542413]	0.501820	3.7946×10^{-7}
11	0.029339	[0.029190, 0.029488]	0.116303	3.7946×10^{-7}

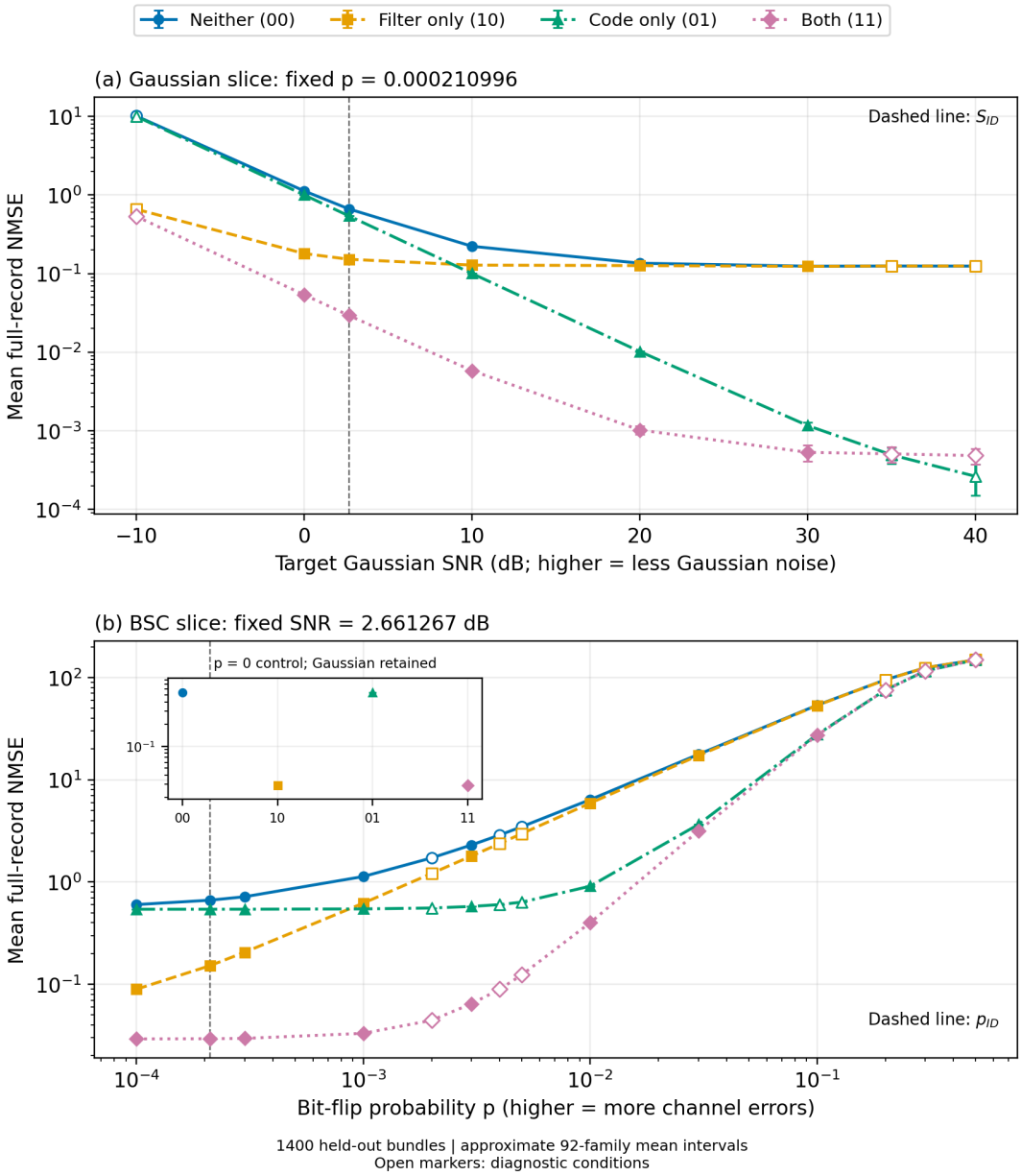


Figure 14: Four configurations, 1400 bundles, approximate 92-family mean intervals. Positive p: log axis; inset: $p = 0$ with Gaussian retained. Filled/open: main/diagnostic points; lines join tested points only.

What the curves establish. At the anchor point, neither/filter-only/code-only/both mean NMSEs are 0.664309, 0.151933, 0.541989 and 0.029339. Both gives the lowest error: coding alone leaves most Gaussian error; filtering alone leaves unprotected bit corruption. As SNR rises, the uncoded BSC floor persists. At 40 dB code-only/both means are $2.6258\times10^{-4}/4.8230\times10^{-4}$; the estimated filter reduction -2.1972×10^{-4} has conservative interval $[-4.4259\times10^{-4}, 3.1494\times10^{-6}]$, including zero. The apparent reversal is therefore not established by the primary rule. Separate clean controls do establish FIR distortion: NMSE 5.7571×10^{-7} without filtering and 3.1484×10^{-4} with it.

The conditional contrasts above inherit conservative bounds from the joint mean event (Appendix B.7); paired pointwise intervals are separate exploratory results. Nonadditivity is retained. At $p = 0.5$ independently fair recovered labels predict NMSE 148.960954 for every configuration,

Table 14: Conditional NMSE reductions and interaction at the anchor/high-SNR points. Positive reductions improve quality; conservative intervals retain negative and unresolved effects.

Condition	Contrast	Mean reduction	Conservative interval
Personal	$G_{F\|C=0}$	5.1238×10^{-1}	$[5.0549 \times 10^{-1}, 5.1926 \times 10^{-1}]$
Personal	$G_{F\|C=1}$	5.1265×10^{-1}	$[5.1208 \times 10^{-1}, 5.1322 \times 10^{-1}]$
Personal	$G_{C\|F=0}$	1.2232×10^{-1}	$[1.1847 \times 10^{-1}, 1.2618 \times 10^{-1}]$
Personal	$G_{C\|F=1}$	1.2259×10^{-1}	$[1.1899 \times 10^{-1}, 1.2620 \times 10^{-1}]$
Personal	I	-2.7385×10^{-4}	$[-7.7348 \times 10^{-3}, 7.1871 \times 10^{-3}]$
40 dB	$G_{F\|C=0}$	-2.1906×10^{-4}	$[-7.1277 \times 10^{-3}, 6.6896 \times 10^{-3}]$
40 dB	$G_{F\|C=1}$	-2.1972×10^{-4}	$[-4.4259 \times 10^{-4}, 3.1494 \times 10^{-6}]$
40 dB	$G_{C\|F=0}$	1.2422×10^{-1}	$[1.2065 \times 10^{-1}, 1.2778 \times 10^{-1}]$
40 dB	$G_{C\|F=1}$	1.2422×10^{-1}	$[1.2065 \times 10^{-1}, 1.2778 \times 10^{-1}]$
40 dB	I	6.6129×10^{-7}	$[-7.1309 \times 10^{-3}, 7.1322 \times 10^{-3}]$

and all four measured intervals contain it. This is the random-label model of Appendix B.4, not small quantisation noise $\Delta^2/12$.

Table 15: Twelve-condition equal-weight objective from 1400 independent bundle scores. Intervals belong to the 92-statistic family; weights are design choices, not channel frequencies.

Config.	$\bar{J}$	Adjusted mean interval	Half-width (%)	kbit/s
00	7.06272	[7.05624, 7.06919]	0.09	224
10	6.65633	[6.64974, 6.66293]	0.10	224
01	3.00505	[3.00105, 3.00905]	0.13	392
11	2.58806	[2.58400, 2.59211]	0.16	392

The objective summarises only its declared 12-condition domain. Using Eq. (B.74), adjacent objective differences $J_{00} - J_{10}$, $J_{10} - J_{01}$ and $J_{01} - J_{11}$ have conservative outward-rounded intervals $[0.39331, 0.41946]$, $[3.64069, 3.66188]$ and $[0.40893, 0.42506]$, respectively. All are positive on the approximate joint mean event, supporting $J_{11} < J_{01} < J_{10} < J_{00}$ for these weights. This weighted-average ranking does not assert the same ordering at every SNR/p, or in a different channel population.

Table 16: Configuration 11 full-grid component powers, including small terms, plus signed cross terms. Sampling-scale residuals here are numerical, not measurable aliasing.

Term (configuration 11)	Personal point	40 dB, personal p
Sampling	1.7504×10^{-25}	1.7504×10^{-25}
Filter	2.2951×10^{-4}	2.2951×10^{-4}
Gaussian	2.1061×10^{-2}	3.8852×10^{-6}
Quantisation	4.0242×10^{-7}	4.0240×10^{-7}
Channel	1.2600×10^{-4}	1.1830×10^{-4}
Signed sum of 10 cross terms	2.2131×10^{-7}	-2.0212×10^{-8}
Total MSE	2.1418×10^{-2}	3.5208×10^{-4}

The five full-grid powers and ten signed cross terms sum to total MSE (Eqs. (B.64), (B.65)). Sampling residual, finite-FIR distortion, Gaussian error, quantisation and channel-induced waveform change remain distinct; they are not forced into a 100 % responsibility chart. Earlier R1/R3 comparisons support the fixed sampling/bit-width decisions; the four switches alone do not identify their separate causal effects. MAE is retained in the personal table to reveal sensitivity to large corruption.

Precision and numerical limits. Seven of 92 means miss the pilot's 5% relative-half-width target, reaching 42.43%. They are coded cases at 20–40 dB and personal p: rare uncorrectable blocks had variance underestimated by the 20-bundle pilot. The frozen 1400 count is retained without optional stopping. These percentages describe uncertainty in mean NMSE, not reconstruction error; 5% is a planning tolerance. At 40 dB the coded filter contrast crosses zero (Table 14), leaving its sign unresolved. Personal comparisons and the four objective means have separate reported intervals; all objective means meet the target, but the positive contrast bounds, not target compliance, support their scoped ranking. Computational agreement does not remove finite-sample uncertainty.

Table 17: Four largest achieved relative half-widths. Joint IDs 19/20/21 are 30/35/40 dB at personal p; the 5 % planning target is not an achieved guarantee.

Statistic (joint ID, config.)	Half-width (%)	Target (%)
21, 01	42.43	5.00
19, 11	23.56	5.00
20, 01	23.36	5.00
21, 11	23.11	5.00

Alternative-method replay (Section 7.1) checks every 123200 path, all 92 means, hashes and component identities within MSE 4.2633×10^{-14}. Full measurements, paired effects, seed/source records and local examples are in `stage4/r43/results/`; excerpts do not replace complete-record metrics.

6 Stage 5: Critical reflection

6.1 Identify the limiting tested decision

For the declared 12-condition objective and original feasibility rules, the limiting tested decision is retaining 14 bit in the existing sequential natural-binary/Hamming layout. Increasing to 16 bit is the only tested feasible change with a positive adjusted objective-gain interval. Small quantisation power does not imply that this is simply dominant quantisation noise; the experiment tests the complete chain.

R4's unchanged cohort diagnoses main-domain channel power 1.8720×10^{0}, Gaussian power 1.7637×10^{-2} and quantisation power 4.0241×10^{-7}, retaining signed cross terms. This locates loss but does not establish a causal design ranking. I therefore predeclare nine one-decision variants: baseline; 385-tap half-width 60/80 Hz; 257/513 taps at 70 Hz; 16 bit; Hamming(15,11); repetition(5,1); no ECC. Everything else retains the standard formulas, fixed matrices, original signal, range and full-grid metric.

Twenty new study-5 pilot bundles plan 350 new confirmation bundles; 300 fails the frozen rule, 350 passes. Each bundle runs all nine variants at 15 conditions: the 12 main conditions plus three zero-main-weight diagnostics ($p = 0.3, 0.5$ at the anchor SNR, and 40 dB at the personal p). Thus $350 \times 9 \times 15 = 47\,250$ confirmation paths are run; only the 12 main conditions enter the

objective. Eq. (B.80) uses tolerance $\max(0.01, 0.05|\bar{g}_{\text{pilot}}|)$; actual confirmation half-widths meet all frozen targets. The approximate 19-statistic family contains eight main gains, eight personal gains and three personal means for R5.2. Gains use Eq. (B.78); whole bundles are the independent units. R4 and pilot records do not enter these means. Same layouts share masks within a bundle; changed lengths use separate registered layouts. Alternative-method replay (Section 7.1) checks all confirmation paths within MSE 4.2633×10^{-14}; kernels also pass cosine-sum checks.

Table 18: New R5 confirmation: objective and paired gains from 350 bundles, approximate 19-family intervals. Feasibility means original 1 % gain/twofold expansion rules; absolute rates remain visible.

One change	Feasible	kbit/s	Mean J	Paired J reduction [interval]
Half-width 60 Hz	No	392.00	2.58890	1.4624×10^{-3} $[1.3297 \times 10^{-3}, 1.5952 \times 10^{-3}]$
Half-width 80 Hz	No	392.00	2.59431	-3.9526×10^{-3} $[-4.0412 \times 10^{-3}, -3.8640 \times 10^{-3}]$
257 taps	No	392.00	2.61074	-2.0373×10^{-2} $[-2.0651 \times 10^{-2}, -2.0096 \times 10^{-2}]$
513 taps	Yes	392.00	2.59185	-1.4839×10^{-3} $[-1.5551 \times 10^{-3}, -1.4127 \times 10^{-3}]$
16 bit	Yes	448.00	2.42149	1.6887×10^{-1} $[1.5944 \times 10^{-1}, 1.7830 \times 10^{-1}]$
Hamming(15,11)	Yes	305.46	5.01125	-2.4209×10^{0} $[-2.4324 \times 10^{0}, -2.4094 \times 10^{0}]$
Repetition(5,1)	No	1120.00	0.45581	2.1345×10^{0} $[2.1268 \times 10^{0}, 2.1423 \times 10^{0}]$
Uncoded	Yes	224.00	6.65212	-4.0618×10^{0} $[-4.0747 \times 10^{0}, -4.0488 \times 10^{0}]$

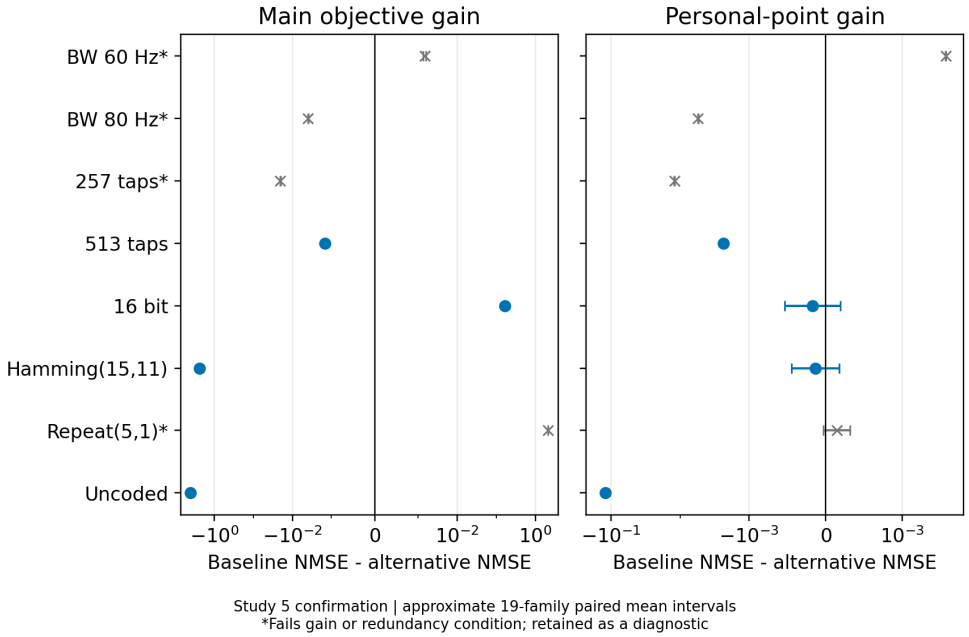


Figure 15: Single-decision gains with 19-family intervals. Grey crosses fail original feasibility rules; repetition cannot be a feasible winner. Symmetric-log panels retain negative/small effects and use distinct ranges.

Baseline objective 2.590362 falls with 16 bit by 0.168868, interval $[0.159440, 0.178295]$, at 392 $\rightarrow$448 kbit/s (14.29 % more traffic). Same-cohort prechannel improvement is only 5.5832×10^{-7}. Most gain is in channel terms, not a 16-fold total-error reduction predicted from $\Delta^2/12$. Eq. (B.37) gives the component ratio; Eq. (B.38) retains joint decoded-bit interactions. Fourteen-bit sample starts alternate at offsets 0/2 modulo 4; 16 bit starts stay aligned. This diagnosis does not isolate alignment from every other consequence of changing bit depth. The anchor-point gain interval includes zero, so no personal improvement is established.

Supplementary check: change packing only. Fix the clean input, 385-tap FIR, 14-bit labels, Hamming(7,4) and 392 kbit/s; Gaussian is absent. Compare sample-major packing with MSB-first bit planes, applying the public inverse permutation after decoding. All 128 BSC patterns per codeword give conditional expected full-grid NMSE (Eq. (B.39), subject to numerical precision). This supplementary diagnostic lies outside the nine-candidate selection and 19-statistic family; only grouping changes, with expected BER fixed.

Table 19: Fixed-14-bit packing: BSC-expected NMSE, Gaussian absent; both layouts send 392 kbit/s. Values are expectations, not confirmation means or confidence intervals.

p	Sample-major	Bit-plane
0	0.000314845	0.000314845
0.000210996	0.000485717	0.000551836
0.01	0.372083	0.517315
0.1	27.2268	36.9535

Results coincide at $p = 0$; bit-plane expected NMSE is larger at all three positive tested probabilities. Grouping therefore affects waveform error, but this layout does not improve it or explain all of the 16-bit gain. Block-outcome/SciPy reconstruction checks the moments; all 16 messages and 128 patterns verify decoder translation invariance. Figure 16 is a shared-mask illustration, not a gain estimate.

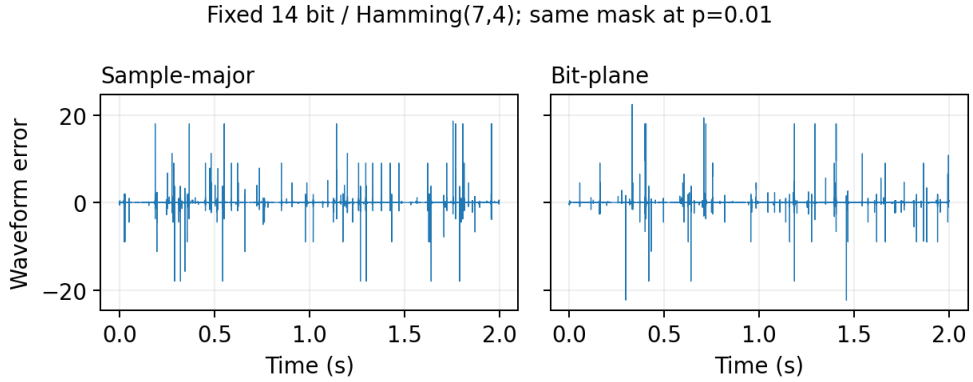


Figure 16: Same-mask waveform errors at $p = 0.01$, Gaussian absent; quantiser, code, rate and axes fixed. One realisation illustrates packing, not expected gain.

Repetition gains 2.134550, interval $[2.126786, 2.142314]$, but sends 1120 kbit/s and violates the original twofold expansion rule. It cannot be a feasible winner. Hamming(15,11) worsens the objective to 5.011250. At $p = 0.1$, exact block-failure laws give 0.1496944/0.4509570/0.0085600 for H74/H15/repetition (Eq. (B.49)); both Hamming codes correct one error, repetition two. Higher code rate is not stronger correction. The 513-tap feasible filter worsens the objective and raises group delay 12 →16 ms; other filter variants fail the original tone-gain guard. No runtime penalty is inferred from their equal FFT length.

This scoped conclusion concerns nine variants and the chosen severity weights, not a global optimum or real-world channel frequencies. Gaussian error dominates personally; FIR distortion matters at 40 dB. The R1–R4 baseline remains frozen; reflection does not retrospectively substitute the 16 bit alternative into its validation.

6.2 Change the code under a 320 kbit/s cap

Under a declared logical cap 320 kbit/s, I choose Hamming(15,11), retaining 16 kHz, 14 bit, $A = 18$, the 385-tap FIR and both anchor parameters. The scenario and its three means were frozen before the R5 pilot; the same 350 confirmation records are reused, not new observations. The cap is a project scenario for transmitting 2 s in 2 s, with rates measured as transmitted bits rather than physical energy. The threshold stays $\tau = 0.057666604564293136$ from R3.4; neither applies to all stress conditions.

Table 20: Exact 2 s costs under the new personal 320 kbit/s cap. Q fit retains the R3 quantisation budget; 11 bit is an analytic cost alternative, without new waveform measurements.

Scheme	Real bits	Pad	Sent bits	kbit/s	Rate fit	Q fit
Hamming(7,4)	448000	0	784000	392.00	No	Yes
Hamming(15,11)	448000	8	610920	305.46	Yes	Yes
Uncoded	448000	0	448000	224.00	Yes	Yes
Repetition(5,1)	448000	0	2240000	1120.00	No	Yes
11 bit + (7,4)	352000	0	616000	308.00	Yes	No

Exact framing (Eq. (B.47)) gives 40728 H15 codewords, eight pad bits and 610920 transmitted bits, 305460 bit/s: 22.08% less traffic. Repetition exceeds the cap; 11 bit/H74 would fit at 308 kbit/s but its existing quantisation-budget ratio $0.354788 > 0.01$ fails the retained requirement. This is an analytic cost alternative, with no invented personal NMSE.

The unchanged acceptance rule requires both rate ≤ 320000 bit/s and the 19-family approximate mean upper bound $\leq \tau$ (Eq. (B.79)). An interval crossing τ is unresolved; this rule is not a guarantee for each record.

Table 21: Same 350 new R5 records at the anchor point. Intervals use the 19-statistic family; perfect payloads require all real information bits, excluding padding.

Scheme	Mean NMSE [interval]	Payload BER	Perfect payloads	Mean rule
Hamming(7,4)	0.029145 [0.028931,0.029358]	3.4439×10^{-7}	319/350	Pass
Hamming(15,11)	0.029279 [0.028995,0.029564]	7.4617×10^{-7}	298/350	Pass
Uncoded	0.151197 [0.145161,0.157233]	2.1117×10^{-4}	0/350	Fail

H15 mean 0.029279, upper bound $0.029564 < \tau$, meets both. H74 meets quality but exceeds the cap; uncoded fits rate but its mean 0.151197 has lower bound above τ. Both Hamming codes have $d_{\min} = 3, t = 1$. The gain in feasibility costs exact-payload reliability: perfect records fall 319/350 $\rightarrow$ 298/350 despite acceptable mean waveform quality. Eq. (B.56) includes the H15 last word's three real/eight padding bits; requiring correct padding would test the wrong event. Unpadded H74 reduces to $(1 - q_{7,1})^{112000}$; uncoded to $(1 - p)^{448000}$. Pointwise Clopper–Pearson perfect-record intervals are separate from the 19-family mean intervals (Eq. (B.71)). Independent checks cover 1050 reused paths, 32898 error patterns and exact costs. H15's worse main objective 5.011250 means this personal cap-driven decision does not replace H74 for the original domain.

7 Stage 6: Integrity and reproducibility

7.1 Reproduce inputs, measurements and report

The verified stack is Python 3.13.5, NumPy 2.1.3, SciPy 1.15.3, Matplotlib 3.10.0 and Tectonic 0.17.0. The explicit Windows dependency lock records the numerical and rendering environment. From the repository root:

```
conda create -n signal-transmission --file conda-win-64-explicit.txt
conda activate signal-transmission
python run_all.py --tectonic PATH_TO_TECTONIC
```

`-data-only` skips PDF compilation. Worker settings and the four-thread parameter scan are documented in `README.md`.

The entry point checks the NPY/WAV input hashes, then regenerates measured CSV, figures and LaTeX fragments. It verifies 316 frozen scientific targets plus seven packing-diagnostic products. CSV fields, arrays, payloads and generated formula/table text are compared exactly; raster comparisons use exact pixels. SVG comparisons retain geometry, visible text and styling while excluding nonvisual export metadata and consistently renamed opaque identifiers. The targets were fixed before packaging and are never refreshed to conceal a mismatch. This release uses the existing validated measurements; publication checks do not constitute a new complete numerical replay.

Computational cross-checks. Alternative methods use direct convolution, midpoint and threshold quantisation, XOR parity, correction-sphere lookup decoding and SciPy resampling where applicable. Candidate kernels also have cosine-series checks. These reproduce existing trials and do not add independent observations. Complete records, rather than individual samples, are the independent units in the reported uncertainty calculations.

The six final vector figures are regenerated automatically from measured CSV by `reproduction/prepare_publication_figures.py`. A report-only build uses `reproduction/build_pdf.py`. Two independent source-only builds with a fixed `SOURCE_DATE_EPOCH` check deterministic PDF output. Runtime evidence and scientific product hashes are saved under `stage6/results/`. Exact replay is scoped to the recorded Windows stack; execution on a second machine has not been tested.

Conclusions and limitations

For the supplied periodic multitone, the selected 16 kHz sampler retains all identified components. The frozen 385-tap FIR, 14-bit quantiser and Hamming(7,4) jointly give the lowest anchor-point mean NMSE, 0.029339, and the lowest declared main-domain objective among the four configurations. These are conditional results for the tested signal, degradation models and severity weights. Increasing to 16 bits improves the new R5 objective at greater traffic cost, while the separate 320 kbit/s anchor-point constraint favours Hamming(15,11) with reduced perfect-payload reliability.

Statistical precision remains a limitation: seven of the 92 R4 means miss the planned 5% relative half-width, reaching 42.43%. The 40 dB coded filter contrast includes zero and does not establish a local ordering. Computational agreement does not remove this sampling uncertainty. The reported intervals, high-SNR FIR distortion and finite resource budgets delimit the conclusions; neither universal improvement nor exact recovery of every payload is claimed.

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Appendix A

Selected parameters and evidence

The frozen validation baseline is collected below; R5 alternatives retain their own constraints (Sections 6.1–6.2). Full RNG trial paths are in `metadata/randomness_design.json`.

Table 22: Frozen baseline and selection evidence. SNR/p mappings are declared project choices, not a supplied external conversion.

Parameter	Selected value	Evidence
Source / sampling	48 kHz, 96000 samples $\rightarrow$ 16000 Hz, $32,000$ samples; 2 s; unmodified NPY units	Section 2.1; Tables 1, 2
RNG / purpose streams	Root 2470964; PCG64; SeedSequence pool size 4; parameters 0, Gaussian 1, BSC 2	Anchor-seed convention; Sections 3.1, 4.3, 5.2
Personal degradation	SNR 2.661267333 dB from $[0, 30]$ dB; $p_{\mathrm{ID}} = 0.00021099569319078592$ from $[10^{-4}, 0.1)$	Eq. (17); Table 10
FIR centres	440, 1320, 2200, 6000 Hz	Tables 1, 5
FIR design	385 symmetric Hann taps; centre ± 70 Hz and conjugate negative bands	Table 6; Figures 6, 7
FIR implementation / delay	Zero-extended linear FFT convolution, $K = 32768$, fixed centre crop; 192 samples, 12 ms causal delay	Eq. (B.20); Sections 3.2–3.3
Quantiser / rounding	$B = 14$, $A = 18$, 16384 endpoint levels; $\Delta = 2.1974 \times 10^{-3}$; shifted round-to-even, $\epsilon = 10^{-9}$, saturation	Eq. (24); Table 8; exhaustive label checks
Bit budget / packing	Quantisation component at most 1% of upstream error power; unsigned natural binary, sample-major, MSB first	Eq. (29); Section 4.1; label round trips
ECC / payload / rate	Systematic Hamming(7,4), $d_{\min} = 3$, $t = 1$; 448000 information / 784000 sent bits; no padding; 392 kbit/s	Eq. (B.43); Table 9; Eq. (B.47)
Reconstruction	Periodic Fourier interpolation to all 96000 original positions	Eq. (B.12); R1 and cross-grid checks
Primary metric	Whole-record NMSE; reference power 0.73	Section 5.1; Figure 13
Quality threshold	$\tau = 0.05766660$; fixed before validation	Eq. (34); Table 11

Appendix B

Formula derivations and implementation conventions

Shared formulas are derived once, with explicit assumptions and published technical references. Finite-array consequences are derived here; statistical methods and design preferences are supplementary. Symbols are local: N is a sampled record length, L the reference/reconstruction length, K a transform length and $L_q = 2^B$ the number of quantiser levels. Powers specify their grid. Receiver algorithms use no reference waveform or true channel errors.

B.1 Finite-record Fourier, windows and power

DFT pair and one-sided scaling. Use the unscaled forward transform and $1/L$ inverse:

$$X[k] = \sum_{n=0}^{L-1} x[n]e^{-j2\pi kn/L}, \quad f_k = \frac{kF_0}{L}, \quad \Delta f = \frac{F_0}{L} = 0.5\,\text{Hz}, \tag{B.1}$$

$$x[n] = \frac{1}{L}\sum_{k=0}^{L-1} X[k]e^{j2\pi kn/L}. \tag{B.2}$$

Euler's cosine pair, $A\cos\theta = (A/2)(e^{j\theta}+e^{-j\theta})$, gives, for an isolated coherent interior component,

$$x[n] = A\cos(2\pi k_0 n/L + \phi) \quad \Longrightarrow \quad X[k_0] = \tfrac{LA}{2}e^{j\phi}. \tag{B.3}$$

For N actual samples and even $K \geq N$, the general windowed scales are

$$\begin{aligned} Y_{a,K}[k] &= \sum_{n=0}^{N-1} y[n]a[n]e^{-j2\pi kn/K}, \quad f_k = \frac{kf_s}{K}, \\ A_{a,K}[k] &= \frac{c_k|Y_{a,K}[k]|}{\sum_n a[n]}, \quad \widehat{S}_{a,K}[k] = \frac{c_k|Y_{a,K}[k]|^2}{f_s\sum_n a[n]^2}. \end{aligned} \tag{B.4}$$

Here $c_k = 1$ at DC/Nyquist and 2 at other positive bins. For $N = K = L$, write $X_w = Y_{w,L}$ and $CG = L^{-1}\sum_n w[n]$; thus $\widehat{A}_w = c_k|X_w|/(L\,CG)$. With $W = \text{DFT}_L(w)$, substitution of the cosine pair yields

$$X_w[k_0] = \tfrac{A}{2}e^{j\phi}W[0] + \tfrac{A}{2}e^{-j\phi}W[2k_0], \qquad W[0] = L\,CG. \tag{B.5}$$

Normalising the first term does not remove mirror, off-bin or overlapping components. Padding changes $\Delta f_K = f_s/K$, not the window-sum denominators.

Orthogonality, Parseval and PSD. For DFT indices $k, \ell \in \{0,\ldots,L-1\}$, the geometric ratio $r = e^{j2\pi(k-\ell)/L}$ gives $(1-r^L)/(1-r) = 0$ when $k \neq \ell$. Insert the IDFT, cancel unequal-index terms, then pair conjugate bins:

$$\sum_{n=0}^{L-1} e^{j2\pi(k-\ell)n/L} = L\delta_{k\ell}, \tag{B.6}$$

$$\sum_n |x[n]|^2 = \frac{1}{L^2}\sum_{k,\ell} X[k]X^*[\ell]L\delta_{k\ell} = \frac{1}{L}\sum_k |X[k]|^2, \tag{B.7}$$

$$P_k = c_k\frac{|X[k]|^2}{L^2}, \qquad \widehat{P} = L^{-1}\sum_n x[n]^2 = \sum_{k=0}^{L/2} P_k. \tag{B.8}$$

Applying the same identity to the padded windowed record gives

$$\sum_{k=0}^{K/2} \widehat{S}_{a,K}[k]\frac{f_s}{K} = \frac{\sum_{k=0}^{K-1}|Y_{a,K}[k]|^2}{K\sum_n a[n]^2} = \frac{\sum_n y[n]^2a[n]^2}{\sum_n a[n]^2}. \tag{B.9}$$

PSD has squared NPY-amplitude/Hz units. Its weighted integral need not equal the raw signal power used for noise generation.

Shared symmetric windows. For length M, $0 \le n < M$, the tested windows are [10, 11]

$$\theta_n = \frac{2\pi n}{M-1}, \quad w_{\text{rect}} = 1, \quad w_{\text{Hann}} = \tfrac{1}{2}(1-\cos\theta_n), \quad w_{\text{Hamming}} = 0.54 - 0.46\cos\theta_n. \tag{B.10}$$

The spectral Hann w_H uses $M = L$; kernel tapers use the FIR length.

B.2 Sampling, interpolation and aliasing

Folding and periodic recovery. For decimation $L = MN$, substitute $x_{\text{ref}}[Mm]$ from the IDFT. Orthogonality retains only $r = k + qN$, including phase:

$$\frac{Y[k]}{N} = \frac{1}{LN}\sum_{r=0}^{L-1} X[r] \sum_{m=0}^{N-1} e^{j2\pi(r-k)m/N} = \sum_{q=0}^{M-1} \frac{X[k+qN]}{L}, \quad 0 \le k < N. \tag{B.11}$$

For real even-N data, $Y = \text{DFT}_N(x_M)$. Pairing the reference IDFT and evaluating at $t/T = n/L$ gives the periodic interpolator:

$$\widehat{x}[n] = \frac{1}{N}\left[Y[0] + 2\,\text{Re}\sum_{k=1}^{N/2-1} Y[k]e^{j2\pi kn/L} + Y[N/2]\cos\left(\frac{\pi Nn}{L}\right)\right]. \tag{B.12}$$

For `irfft(Z,L)`, $L > N$, the old Nyquist coefficient splits into an interior conjugate pair, requiring

$$Z[k] = \frac{L}{N}Y[k]\ (k < N/2), \quad Z[N/2] = \frac{L}{2N}Y[N/2], \quad Z[k] = 0\ (k > N/2). \tag{B.13}$$

For $L = N$ it is not split. DC/phased/Nyquist tests and independent SciPy resampling check the convention [12, 19]. The measured two-second periodicity supports this ideal-baseband recovery. Its inputs are only the samples and target length; it does not recover arbitrary nonperiodic records.

Boundary phase and signed aliases. At $f_0 = f_s/2$, the unknown sine component is lost:

$$A\cos(\pi m + \phi) = A(-1)^m\cos\phi. \tag{B.14}$$

For nonboundary $f_d = f_0 - qf_s \in (-f_s/2, f_s/2)$, $f_d \neq 0$, and $f_a = |f_d|$,

$$A\cos(2\pi f_0 m/f_s + \phi) = A\cos(2\pi f_a m/f_s + \phi_a), \qquad \phi_a = \text{sgn}(f_d)\phi. \tag{B.15}$$

In the 8 kHz diagnostic, the retained low-frequency terms cancel, leaving

$$e_8(t) \simeq -0.1\sin(2\pi\,2000t) - 0.1\sin(2\pi\,6000t). \tag{B.16}$$

These distinct integer-bin tones are orthogonal on the common grid, so

$$\text{MSE}_{8,\text{pred}} = 0.1^2/2 + 0.1^2/2 = 0.01. \tag{B.17}$$

This explains measured phases after the experiment; no source tones enter the recovery algorithm.

B.3 Linear filtering, finite boundaries and reconstruction energy

Convolution and padding. The reference impulse expansion $y[n] = \sum_m y[m]\delta[n-m]$, linearity and time invariance give

$$z[n] = \mathcal{T}\{y\}[n] = \sum_m y[m]\mathcal{T}\{\delta[\cdot - m]\}[n] = \sum_m y[m]h[n-m]. \tag{B.18}$$

For circular convolution, $r = (n-m) \bmod K$ separates the DFT sum:

$$Z_c[k] = \sum_m h[m]e^{-j2\pi km/K}\sum_r y[r]e^{-j2\pi kr/K} = H_K[k]Y_K[k]. \tag{B.19}$$

Zero extension gives $N + M - 1$ full outputs. Linear convolution is therefore

$$z_K[n] = \frac{1}{K} \sum_{k=0}^{K-1} H_K[k] Y_K[k] e^{j2\pi kn/K}, \qquad K \geq N + M - 1. \tag{B.20}$$

retaining the first $N + M - 1$ values, with zero padding. The 513-tap prototype has 32512 full outputs; $K = 32768$ is convenient, not a NumPy requirement. An insufficient-padding counterexample is

$$[1, 2, 1] * [2, 0, 2] = [2, 4, 4, 4, 2], \qquad \mathrm{IDFT}_4(Y_4 H_4) = [4, 4, 4, 4]. \tag{B.21}$$

A centred three-point mean has aligned response

$$h = [1, 1, 1]/3, \quad d = 1, \qquad G_3(\omega) = e^{j\omega d} \sum_{m=0}^{2} h[m] e^{-j\omega m} = \frac{1 + 2\cos\omega}{3}. \tag{B.22}$$

One finite-IDFT filter family. For signed frequencies f_k, measured corrupted-spectrum centres f_i, half-width b, odd M and $d = (M - 1)/2$,

$$B_b[k] = \mathbf{1}\Big\{ \min_i |f_k| - f_i| \leq b \Big\}, \tag{B.23}$$

$$h_{M,b,w}[m] = w[m] \frac{1}{N} \sum_{k=0}^{N-1} B_b[k] e^{j2\pi k(m-d)/N}, \qquad d = (M - 1)/2. \tag{B.24}$$

Use the symmetric windows in Eq. (B.10) on the kernel, $0 \leq m < M$. The Hann prototypes use 513 taps and 80 Hz half-width in R2.1, then 385 taps and 70 Hz in R2.2. The low-pass comparator uses $B_b = \mathbf{1}_{\{|f_k| \leq 6600\,\mathrm{Hz}\}}$. This project construction uses the declared R1 priors, not true noise. The realised response is $\mathrm{DFT}_K(h)$, not the binary template.

With zero extension, the reference-independent crop is

$$\widehat{x}[n] = z_{\mathrm{full}}[n + d], \quad 0 \leq n < N. \tag{B.25}$$

The offline crop uses future samples; causal FIR delay is d/f_s (16 ms for the prototype), distinct from whole-record FFT buffering.

Finite-boundary noise gain, distortion and crossover. For a fixed filter, write $d_x = F(x) - x$, $v = F(e)$ and $y = x + e$:

$$F(y) - x = d_x + v, \qquad \mathrm{MSE} = (\|d_x\|^2 + \|v\|^2 + 2d_x^{\mathsf{T}} v)/N. \tag{B.26}$$

For new independent zero-mean noise, $g[\ell] = h[\ell + d]$, $d < N$, each lag overlaps $N - |\ell|$ positions. Thus

$$G_{\mathrm{finite}} = \sum_{\ell=-d}^{d} g[\ell]^2 (1 - |\ell|/N),$$

$$\mathbb{E}[\mathrm{MSE}] = D + \sigma^2 G_{\mathrm{finite}}, \qquad D = \|d_x\|^2/N. \tag{B.27}$$

The cross term vanishes in expectation, not in each trial. For $D > 0, G < 1$, $D + \sigma^2 G = \sigma^2$ gives $\sigma^2 = D/(1 - G)$ and

$$S_* = 10 \log_{10} \frac{P_x(1 - G_{\mathrm{finite}})}{D} \approx 34.79 \text{ dB}. \tag{B.28}$$

Reconstruction-grid energy and expected noise. For the frozen filter, $D = 2.2951 \times 10^{-4}$ and $G = 0.05322427$ in Eq. (B.27). Let $V = \mathrm{DFT}_N(v)$ for a real even-N error vector and

$A_\pi = V[N/2]/N = N^{-1}\sum_n (-1)^n v[n]$. For interpolation R to $L = 3N$, ordinary pairs retain their mean-square contribution; splitting the old Nyquist coefficient halves its power:

$$\begin{aligned}\frac{1}{N}\|v\|^2 &= \frac{|V[0]|^2 + 2\sum_{k=1}^{N/2-1}|V[k]|^2 + |V[N/2]|^2}{N^2},\\ \frac{1}{L}\|R(v)\|^2 &= \frac{|V[0]|^2 + 2\sum_{k=1}^{N/2-1}|V[k]|^2 + \frac{1}{2}|V[N/2]|^2}{N^2}\\ &= \frac{1}{N}\|v\|^2 - \frac{1}{2}A_\pi^2. \end{aligned} \tag{B.29}$$

With $a_n = (-1)^n/N$, independence gives $\mathbb{E}[(a^\mathsf{T}Fe)^2] = \sigma^2\|F^\mathsf{T}a\|^2$, so

$$G_{48} = G - \frac{1}{2}\|F^\mathsf{T}a\|^2 = 0.05322427093, \qquad G_{0,48} = 1 - \frac{1}{2N} = 0.999984375. \tag{B.30}$$

The 48 kHz predictions are $D_{48} + \sigma^2 G_{48}$ and $\sigma^2 G_{0,48}$, with measured clean distortion $D_{48} = 2.2951 \times 10^{-4}$; negligible clean interpolation residuals are omitted. Actual errors use the original reference, rather than assuming the two grids are equivalent.

Independent finite cosine-series kernel. For disjoint interior bands centred on DFT bins, integer $J_b = bN/f_s$ and $C_b = 2J_b + 1$, the finite geometric series gives an independent check:

$$\begin{aligned} S_{C_b}(\ell) &= \sum_{q=-J_b}^{J_b} e^{j2\pi q\ell/N} = \begin{cases} C_b, & \ell = 0,\\ \dfrac{\sin(C_b\pi\ell/N)}{\sin(\pi\ell/N)}, & \ell \neq 0, \end{cases}\\ h[m] &= \frac{2w[m]}{N} S_{C_b}(m-d)\sum_i \cos\!\left(\frac{2\pi f_i(m-d)}{f_s}\right). \end{aligned} \tag{B.31}$$

B.4 Quantisation, binary labels and amplitude errors

Endpoint codebook and shifted rounding. The endpoint mapping, Eq. (24), has spacing $\Delta = 2A/(L_q - 1)$ versus the ideal-model $2A/L_q$. The normalised epsilon shifts rounding by $A\epsilon$, giving $|z_q - z_c| \le \Delta/2 + A\epsilon$ apart from floating-point effects. The even endpoint codebook omits zero; the retained shift maps zero to $-\Delta/2$. With clipping, $e_q = (z_c - z) + (z_q - z_c)$, whose power includes their cross term. Actual errors need not be white, independent or zero-mean [24].

Common headroom and an overload bound. Both branches share a range fixed before noise generation for the -10 to 40 dB stress envelope. Let $M_b = \max_n |F_b(x)[n]|$, $g_0 = 1$, $g_F = \|h\|_2$ and $\sigma_{\max} = \sqrt{P_x 10^{-S_{\min}/10}} = 2.701851$ at $S_{\min} = -10$ dB. Independence gives $\mathrm{Var}[(F\eta)[n]] \le \sigma^2\|h\|_2^2$, including boundaries. Six-sigma headroom yields

$$A = \left\lceil \max_b (M_b + 6\sigma_{\max} g_b) \right\rceil = \lceil \max(17.56984, 5.11071) \rceil = 18. \tag{B.32}$$

For unbounded Gaussian noise this is not a zero-overload guarantee: $\Pr(\text{any overload}) \le 2N\,\mathrm{erfc}(6/\sqrt{2}) = 1.2628 \times 10^{-4}$ for the two $N = 32000$ branches by the union bound, without branch independence. Overloads are counted before clipping; the stress envelope does not expand R2's design claims.

Uniform-error integral and SQNR conventions. The reference's unoverloaded model has $p_e(e) = 1/\Delta$ on $[-\Delta/2, \Delta/2)$ and $\mathbb{E}[e] = 0$ by symmetry. Its model variance is [1]

$$\mathbb{E}[e^2] = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} e^2\, de = \frac{1}{\Delta}\left[\frac{e^3}{3}\right]_{-\Delta/2}^{\Delta/2} = \frac{\Delta^2}{12}. \tag{B.33}$$

For endpoint spacing and $R_a = 2A$, this gives the model

$$\mathrm{SQNR}_{\text{model}} = 10\log_{10}\!\left(\frac{12P_z}{\Delta^2}\right) = 10\log_{10}\!\left(\frac{12P_z(2^B-1)^2}{R_a^2}\right). \tag{B.34}$$

A full-range sine has $P_z = R_a^2/8$, so the ideal uniform spacing gives $10\log_{10}(1.5\,4^B) = 10\log_{10}1.5 + 20B\log_{10}2 \simeq 1.76 + 6.02B$ dB. Endpoint spacing adds $20\log_{10}(1-2^{-B})$. Neither sine formula is the actual multitone/noisy-input SQNR.

Random-label power and the p=0.5 full-grid check. At $p = 0.5$, received bits are independent fair bits, independent of the payload. Hamming(7,4)'s 16 equal eight-word correction spheres give uniform decoded messages, hence independent fair information bits and uniform disjoint 14-bit labels. For uniform $j \in \{0, \ldots, L_q - 1\}$, $\mathbb{E}[j] = (L_q-1)/2$ and $\mathrm{Var}(j) = (L_q^2-1)/12$, giving

$$V_q = \left(\frac{2A}{L_q - 1}\right)^2 \mathrm{Var}(j) = \frac{A^2(L_q+1)}{3(L_q-1)}. \tag{B.35}$$

Here $A = 18, L_q = 2^{14}$. This is decoded-level power, not $\Delta^2/12$. The reference cross term has zero expectation; Nyquist correction gives

$$\mathbb{E}[d \mid p = 0.5] = 1 + \frac{V_q}{P_{\mathrm{ref}}}\left(1 - \frac{1}{2N}\right) = 148.960954, \qquad N = 32000. \tag{B.36}$$

Bit depth, label significance and code-block coupling. The endpoint model predicts only the quantisation-component ratio

$$\frac{P_{q,16}}{P_{q,14}} = \left(\frac{2^{14}-1}{2^{16}-1}\right)^2 \simeq \frac{1}{16}, \tag{B.37}$$

For true MSB-first bits $b_{n,u}$ and error indicators $w_{n,u}$, XOR is $\widehat{b}_{n,u} = b_{n,u} + (1-2b_{n,u})w_{n,u}$. Therefore

$$\begin{aligned} \ell[n] &= \sum_{u=0}^{B-1} 2^{B-1-u} b_{n,u}, \quad z_q[n] = -A + \Delta_B \ell[n], \\ \delta z[n] &= \Delta_B(\widehat{\ell}[n] - \ell[n]) = \Delta_B \sum_{u=0}^{B-1} (1-2b_{n,u}) 2^{B-1-u} w_{n,u}. \end{aligned} \tag{B.38}$$

At $B = 14$, one MSB error has 2^{13} times an LSB error's amplitude magnitude and 2^{26} times its squared magnitude. Squaring retains joint $w_{n,u}w_{n,v}$ terms. Label starts $(Bn) \bmod 4$ alternate 0/2 for 14 bits but stay 0 for 16; dependent within-codeword errors therefore interact differently with significant bits. Component/cross-term measurements capture the combined change, not an isolated alignment effect or resolution-only gain.

Fixed-resolution packing diagnostic. With fixed 14-bit labels, $\alpha_{n,u} = \Delta 2^{13-u}(1-2b_{n,u})$. Let A_b place codeword b's four slopes in their sample positions: $\delta z = \sum_b A_b w_b$. The 128 BSC patterns give residual-bit mean q and covariance C. Independent blocks give $m = \mathbb{E}[\delta z] = \sum_b A_b q$ and $\Sigma = \mathrm{Cov}(\delta z) = \sum_b A_b C A_b^T$. Apply Eq. (B.29) to the zero-mean fluctuation:

$$\mathbb{E}[d] = \frac{1}{P_{\mathrm{ref}}}\left\{\frac{\|\mathcal{R}(z_q + m) - x_{\mathrm{ref}}\|^2}{L} + \frac{\mathrm{tr}\,\Sigma}{N} - \frac{a^T \Sigma a}{2N^2}\right\}, \quad a_n = (-1)^n. \tag{B.39}$$

The final term corrects split Nyquist energy. Packing changes A_b, with q, C, z_q fixed; Gaussian is absent. The code accumulates block moments without storing Σ.

B.5 Systematic ECC, channel error laws and payload counting

Counting fixed-width samples gives

$$R_{\mathrm{raw}} = Bf_s, \qquad N_{\mathrm{bits}} = BN, \qquad \frac{R_{\mathrm{raw},i}}{R_{\mathrm{raw},16k}} = \frac{f_{s,i}}{16000}. \tag{B.40}$$

Fixed matrices, syndrome, distance and miscorrection. The reference distance and correction radius are

$$d_H(\mathbf{u},\mathbf{v}) = \sum_{i=0}^{n-1} \mathbb{1}\{u_i \neq v_i\}, \qquad d_{\min} = \min_{\mathbf{u}\neq\mathbf{v}\in C} d_H(\mathbf{u},\mathbf{v}), \tag{B.41}$$

$$R = \frac{k}{n}, \qquad t = \left\lfloor \frac{d_{\min}-1}{2} \right\rfloor. \tag{B.42}$$

A validity check detects up to $d_{\min}-1$ errors; correction guarantees only t. All arithmetic is in GF(2), using integer NumPy matrix multiplication modulo two over GF(2). The actual row-message matrices are

$$G_7 = \left[\begin{array}{cccc|ccc} 1&0&0&0&1&1&0\\ 0&1&0&0&1&0&1\\ 0&0&1&0&0&1&1\\ 0&0&0&1&1&1&1 \end{array}\right], \qquad H_7 = \left[\begin{array}{cccc|ccc} 1&1&0&1&1&0&0\\ 1&0&1&1&0&1&0\\ 0&1&1&1&0&0&1 \end{array}\right]. \tag{B.43}$$

For $G = [I_k \mid P]$ and $H = [P^T \mid I_{n-k}]$,

$$GH^T = P + P = 0, \quad \mathbf{c} = \mathbf{m}G, \quad \mathbf{r} = \mathbf{c} + \mathbf{e}, \quad \mathbf{s} = \mathbf{r}H^T = \mathbf{e}H^T. \tag{B.44}$$

A single-error syndrome equals the corresponding column of H. Its columns, in order, are 110, 101, 011, 111, 100, 010, 001. Use column matching rather than binary-syndrome indexing; zero syndrome leaves the word unchanged. Extract the first k bits after correction. The reference example $1100 \mapsto 1100011$ becomes 1110011 after flipping zero-based position 2; syndrome 011 restores it. Enumeration of all 16 messages gives minimum nonzero weight 3. Linearity, $d_H(c_1,c_2) = w_H(c_1+c_2)$, proves $d_{\min} = 3$. The disjoint radius-one spheres contain $2^4(1+7) = 2^7$ words, a perfect partition; sphere size alone is not proof of correction.

Two flips $0000000 \to 1100000$ also give syndrome 011: correction produces the wrong valid word 1110000, message 1110, with final syndrome zero. This decoder is not SECDED; validity does not prove message recovery.

Alternative codes and majority decoding. For the $(15, 11)$ reference code, define $G_{15} = [I_{11} \mid P_{15}]$ and $H_{15} = [P_{15}^T \mid I_4]$ by these ordered four-bit parity rows:

$$\begin{aligned} \text{rows}(P_{15}) = (&1100, 1010, 1001, 0110, 0101, 0011,\\ &1110, 1101, 1011, 0111, 1111). \end{aligned} \tag{B.45}$$

Each string is one binary row. Distinct nonzero columns of H_{15} exclude weight-one/two codewords; the first generator row has weight three. Thus $d_{\min} = 3, t = 1, R = 11/15$. The five-repetition reference alternative has $G_5 = [1\ 1\ 1\ 1\ 1]$, $H_5 = [\mathbf{1}_4 \mid I_4]$, $d_{\min} = 5, t = 2, R = 1/5$. Majority fails on three or more flips:

$$P_{b,5} = 10p^3(1-p)^2 + 5p^4(1-p) + p^5. \tag{B.46}$$

The ideal-model three-repetition example instead gives $3p^2 - 2p^3$. Uncoded $(1, 1)$ has $d_{\min} = 1, t = 0, R = 1, P_b = p$.

Exact framing, padding and effective rate. For N labels of B bits, duration T and an (n, k) code,

$$\begin{aligned} M_b &= NB, \qquad a = (-M_b) \bmod k, \qquad N_c = \lceil M_b/k \rceil,\\ M_t &= nN_c, \qquad R_{\text{eff}} = M_b/M_t, \qquad r_t = M_t/T. \end{aligned} \tag{B.47}$$

Here $N = 32000, B = 14, T = 2$ s, so $M_b = 448000$. R3.2 also uses $M = M_b, v_t = r_t$. The last word has $b = M_b - k(N_c - 1)$ real bits. Padding is encoded and counted in the logical sending rate but excluded from real-payload BER; headers and physical energy lie outside this rate definition.

Independent BSC flips imply

$$E_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p), \quad J_b = \sum_{i=1}^{n} E_i \sim \text{Binomial}(n,p), \quad \Pr(J_b = j) = \binom{n}{j} p^j (1-p)^{n-j}. \tag{B.48}$$

Exact complete-block failure and decoded BER. For the verified perfect Hamming and odd-repetition decoders, complete padded-message failure occurs exactly when error weight exceeds t:

$$q_{n,t}(p) = 1 - \sum_{j=0}^{t} \binom{n}{j} p^j (1-p)^{n-j} = \sum_{j=t+1}^{n} \binom{n}{j} p^j (1-p)^{n-j}. \tag{B.49}$$

The positive tail is numerically stable. This law is not assumed for arbitrary ECC; real-payload BER excludes Hamming(15,11)'s eight padding bits. Translation invariance, Eq. (B.54), permits all-zero message enumeration. Grouping patterns by weight gives

$$\begin{aligned} A_{c,j} &= \sum_{\mathbf{e}:\, w_H(\mathbf{e})=j} w_H(\mathcal{D}_c(\mathbf{e})), \\ P_{b,c}(p) &= \frac{1}{k} \sum_{j=0}^{n} A_{c,j} p^j (1-p)^{n-j}. \end{aligned} \tag{B.50}$$

Multiplicity is included in $A_{c,j}$. The (7,4) coefficients $(0, 0, 36, 76, 64, 48, 28, 4)$ give $P_b = 9p^2 + O(p^3)$; (15,11) gives $21p^2 + O(p^3)$. Also $q_{7,1} = 1 - (1-p)^7 - 7p(1-p)^6 = 21p^2 + O(p^3)$.

Capacity and rate are distinct from finite-code error. The memoryless BSC capacity is

$$H_2(p) = -p \log_2 p - (1-p)\log_2(1-p), \qquad C(p) = 1 - H_2(p), \tag{B.51}$$

Use $0 \log_2 0 = 0$; $C(p) = 4/7$ gives $p = 0.08764841$. Capacity does not predict this finite decoder's BER.

Exact finite-code BER crossover. Expanding the (7,4) enumerator and factoring gives the nontrivial crossover in $(0, 1/2)$:

$$P_{b,7}(p) = 9p^2 - 26p^3 + 30p^4 - 12p^5,$$

$$P_{b,7}(p) - p = -p(2p-1)(p-1)(6p^2 - 6p + 1), \tag{B.52}$$

$$p_{\text{BER}} = (3 - \sqrt{3})/6 = 0.21132487. \tag{B.53}$$

Decoder invariance under a shared mask. Adding a valid codeword leaves syndrome and correction vector $\kappa(\mathbf{s})$ unchanged. For information projection π_4,

$$\mathbf{s} = (\mathbf{c} \oplus \mathbf{e})H^T = \mathbf{e}H^T, \qquad D_1(E_1(\mathbf{b}) \oplus \mathbf{e}) = \mathbf{b} \oplus \pi_4[\mathbf{e} \oplus \kappa(\mathbf{e}H^T)]. \tag{B.54}$$

Hence $\mathcal{D}(c \oplus e) = m \oplus \mathcal{D}(e)$; odd-repetition majority has the same translation property. The same mask gives identical decoded error patterns across messages.

Exact real-payload probability with a partial final word. Let $b = M_b - (N_c - 1)k$ and let $a_{b,w}$ count weight-w patterns whose decoded first b bits remain correct. Translation invariance and independent disjoint BSC words give

$$s_b(p) = \sum_{w=0}^{n} a_{b,w} p^w (1-p)^{n-w}, \tag{B.55}$$

$$P_{\text{exact payload}} = (1 - q_{n,t}(p))^{N_c - 1} s_b(p). \tag{B.56}$$

For (15,11), $b = 3$ and eight bits are padding, so $(1 - q_{15,1})^{40728}$ is not exact real-payload recovery. For unpadded (7,4), $b = 4$, $s_b = 1 - q_{7,1}$, yielding $(1 - q_{7,1})^{112000}$; uncoded transmission gives $(1-p)^{448000}$.

B.6 Reconstruction metrics and error accounting

Common finite-record metrics and units. For reference u, recovery $\widehat{u}$, Q grid positions and $P_u > 0$,

$$a[j] = \widehat{u}[j] - u[j], \quad P_u = Q^{-1}\sum_j u[j]^2, \quad E = \mathrm{MSE} = Q^{-1}\sum_j a[j]^2,$$
$$\mathrm{RMSE} = \sqrt{E}, \qquad \mathrm{NMSE} = E/P_u, \qquad S_{\mathrm{error}} = 10\log_{10}(P_u/E). \tag{B.57}$$

Cauchy–Schwarz gives

$$\mathrm{MAE} = Q^{-1}\sum_j |a[j]| \le \sqrt{Q^{-1}\sum_j a[j]^2} = \mathrm{RMSE}\,. \tag{B.58}$$

MAE is not a monotone transform of MSE.

For paired repeat r,

$$\delta_r = E_{0,r} - E_{F,r}, \qquad g_r = S_{\mathrm{error},F,r} - S_{\mathrm{error},0,r} = 10\log_{10}\frac{E_{0,r}}{E_{F,r}}. \tag{B.59}$$

Boundary diagnostics and the aggregate denominator. For edge set B of size $2q$ and complement I,

$$E_B = \frac{1}{2q}\sum_{j\in B} a[j]^2, \qquad E_I = \frac{1}{Q-2q}\sum_{j\in I} a[j]^2,$$
$$E = \frac{2q}{Q}E_B + \frac{Q-2q}{Q}E_I, \qquad \eta_B = \frac{(2q/Q)\overline{E}_B}{\overline{E}}. \tag{B.60}$$

The aggregate ratio is not the mean of individual edge/total ratios.

Component budget and total error. The squared-sum identity in Eq. (B.65) applies also to upstream error a and extra quantisation q. Define $\rho_q = \mathrm{mean}(q^2)/\,\mathrm{mean}(a^2)$ for nonzero upstream power. Cauchy–Schwarz bounds their cross term, giving

$$\left|\frac{\mathrm{mean}[(a+q)^2]}{\mathrm{mean}(a^2)} - 1\right| \le \rho_q + 2\sqrt{\rho_q}. \tag{B.61}$$

A 1% component budget permits a conservative 21% total-MSE change; the observed largest increase is 0.5840%.

Expanding centred squares with $\bar{e} = L^{-1}\sum_\ell e[\ell]$ gives

$$V_e = L^{-1}\sum_\ell (e[\ell] - \bar{e})^2 = D - \bar{e}^2, \qquad D = V_e + \bar{e}^2. \tag{B.62}$$

Variance alone omits bias. Waveform MSE uses L, not the $L-1$ sample-variance divisor.

For centred reference/recovery arrays x_c, y_c,

$$\rho = \frac{\sum_\ell x_c[\ell]y_c[\ell]}{\sqrt{\sum_\ell x_c[\ell]^2 \sum_\ell y_c[\ell]^2}}. \tag{B.63}$$

Five components and ten signed cross terms. The five full-grid errors are

$$e_s = \mathcal{R}x - x_{\mathrm{ref}}, \qquad e_f = \mathcal{R}F_f x - \mathcal{R}x, \qquad e_g = \mathcal{R}F_f\eta,$$
$$e_q = \mathcal{R}z_q - \mathcal{R}z_f, \qquad e_b = \widehat{x} - \mathcal{R}z_q, \tag{B.64}$$

Expanding their squared sum without independence assumptions gives

$$\widehat{x} - x_{\mathrm{ref}} = \sum_i e_i, \qquad D = \sum_i \frac{\|e_i\|^2}{L} + 2\sum_{i<j}\frac{\langle e_i, e_j\rangle}{L}. \tag{B.65}$$

All five powers and ten signed cross terms are saved; they are not normalised into a 100% responsibility chart.

For repeated complete-record NMSE values,

$$\bar{d} = R^{-1} \sum_r d_r, \qquad S_{\text{err,from mean}} = -10 \log_{10} \bar{d}. \tag{B.66}$$

Generally this differs from averaging per-record dB values. Exact zero error gives $+\infty$; no epsilon floor is inserted.

B.7 Uncertainty, planning and decision rules

One mean/variance/interval convention. For R independent complete-record/bundle values v_r,

$$\bar{v} = R^{-1} \sum_{r=0}^{R-1} v_r, \qquad s_v^2 = (R-1)^{-1} \sum_r (v_r - \bar{v})^2, \tag{B.67}$$

$$[L_v, U_v] = \bar{v} \pm t_{1-\alpha/(2m),R-1} \frac{s_v}{\sqrt{R}}. \tag{B.68}$$

Here $\alpha = 0.05$. Independence gives $\mathrm{Var}(\bar{v}) = \mathrm{Var}(v_r)/R$. These are approximate t/CLT mean intervals for potentially nonnormal records. Bonferroni's union bound $\Pr(\bigcap_{j=1}^m A_j) \geq 1 - \sum_j \Pr(A_j^c)$ needs no independence between statistics, but does not make marginal coverage exact [5, 8, 9, 20, 21]. Applications use $(R, m) = (50, 1)$ in R2.3, $(5, 1)$ in R3.2, $(300, 30)$ in R3.4, $(1400, 92)$ in R4.3 and $(350, 19)$ in R5. Pilot mean/SD estimates use 20 bundles and never enter confirmation means. Paired differences and objectives are formed within bundles first; correlated time positions and bits do not increase R. These statistical methods supplement the DSP reference.

For M_c independent disjoint codewords with failure probability $q(p)$,

$$F \sim \text{Binomial}(M_c, q(p)), \qquad \mathbb{E}[F] = M_c q(p), \qquad \Pr(F = 0) = (1 - q(p))^{M_c}. \tag{B.69}$$

Exact proportion intervals and the zero-event bound. With beta quantile B^{-1}, inversion of the binomial tails $\Pr_L(F \geq f) = 0.025$, $\Pr_U(F \leq f) = 0.025$ gives the pointwise 95% Clopper–Pearson interval [2], including endpoint cases:

$$L = \begin{cases} 0, & f = 0, \\ B^{-1}(0.025; f, M_c - f + 1), & f > 0, \end{cases} \tag{B.70}$$

$$U = \begin{cases} 1, & f = M_c, \\ B^{-1}(0.975; f + 1, M_c - f), & f < M_c. \end{cases} \tag{B.71}$$

For c perfect payloads among R records, substitute $f = c, M_c = R$; the event and independent unit change. SciPy beta quantiles are checked against its exact binomial-test interval [18]. These exact intervals can be conservative [22]. For zero observed failures, $(1 - q)^{M_c} = 0.05$ gives the *one-sided* 95% bound

$$q_U = 1 - 0.05^{1/M_c}; \qquad M_c = 4480000 \implies q_U \simeq 6.6869 \times 10^{-7}. \tag{B.72}$$

Zero observations do not establish a zero underlying rate.

Planning a new record from a nested design bank. R3.4 uses five earlier Gaussian banks with eight BSC masks each. Conditional on Gaussian input Z, write $\mu(Z) = \mathbb{E}[d \mid Z]$ and $v(Z) = \mathrm{Var}(d \mid Z)$. Total variance gives $\mathrm{Var}(d) = \mathbb{E}[v(Z)] + \mathrm{Var}[\mu(Z)]$; bank-mean variance estimates the second term plus $\mathbb{E}[v(Z)]/8$. With within-bank sample variance s_g^2 and mean $\overline{d}_g$, subtract that reduced contribution (truncate at zero) and add the full within-bank variance:

$$\widehat{V} = \overline{s_g^2} + \max\left(s_{\overline{d}_g}^2 - \frac{\overline{s_g^2}}{8}, 0 \right). \tag{B.73}$$

Contrasts and interaction bounds on a joint event. On the joint primary-interval event, conservative contrasts satisfy

$$\mu_a - \mu_b \ \in \ [L_a - U_b,\ U_a - L_b]. \tag{B.74}$$

For $I = \mu_{00} - \mu_{10} - \mu_{01} + \mu_{11}$,

$$\begin{aligned} L_I &= L_{00} - U_{10} - U_{01} + L_{11}, \\ U_I &= U_{00} - L_{10} - L_{01} + U_{11}. \end{aligned} \tag{B.75}$$

For paired $\delta_r = d_{a,r} - d_{b,r}$, expanding centred squares gives

$$s_\delta^2 = s_a^2 + s_b^2 - 2s_{ab}, \qquad s_{ab} = \frac{1}{R-1}\sum_r (d_{a,r} - \bar{d_a})(d_{b,r} - \bar{d_b}). \tag{B.76}$$

Apply Eq. (B.68) to δ with $R = 1400, m = 1$ for pointwise paired intervals, distinct from the joint primary bounds.

Bundle-level paired objectives. The 12 unique main conditions have equal design weights, not inferred channel occurrence probabilities. Define the objective and paired gain once:

$$J_{fc,r} = \frac{1}{12}\sum_{s\in\mathcal{S}_{\text{main}}} d_{fc,s,r}, \quad \bar{J}_{fc} = R_v^{-1}\sum_r J_{fc,r}, \tag{B.77}$$

$$g_{v,r} = J_{\text{base},r} - J_{v,r} = \frac{1}{12}\sum_{s\in\mathcal{S}_{\text{main}}} (d_{\text{base},s,r} - d_{v,s,r}). \tag{B.78}$$

The R5.2 joint constraint. At the prespecified anchor-point scenario, the upper mean bound from Eq. (B.68), with $R = 350, m = 19$, must satisfy

$$U_v \le \tau, \qquad r_{t,v} \le 320000 \text{ bit/s}. \tag{B.79}$$

The quality threshold is unchanged; this is a population-mean decision, not a per-record guarantee.

One pilot precision-planning relation. Using phase-0 standard deviations $\widetilde{s}_j$, choose the smallest allowed count meeting every prespecified half-width target:

$$t_{1-0.05/(2m),R-1}\frac{\widetilde{s}_j}{\sqrt{R}} \le \epsilon_j \quad \text{for all } j, \qquad R \ge 200, \quad R \in 50\mathbb{N}. \tag{B.80}$$

R4 uses $m = 92$, $\epsilon_j = 0.05\widetilde{\mu}_j$ and selects 1400. R5 uses $m = 19$, $\epsilon_j = \max(0.01, 0.05|\widetilde{\mu}_j|)$ and selects 350; 300 fails some pilot targets. R3.4 uses $m = 30$, $\epsilon = 0.1\tau$, $\widetilde{s}_j = \sqrt{\widehat{V}_j}$ for uncoded $p = 0.0001$ and coded $p = 0.003$, selecting 300. Planning does not guarantee realised precision: 7/92 R4 means miss the 5% relative half-width target, with maximum 42.43%. This is uncertainty in estimated means, not reconstruction error. No confirmation data enter planning or trigger extra repeats.

B.8 Parameter-mapping convention

The declared project mapping $p_{\text{ID}} = 10^{-4+3u}$, uniform $u \in [0, 1)$, gives CDF $(\log_{10} p + 4)/3$. Differentiation yields

$$f_{p_{\text{ID}}}(p) = \frac{1}{3\ln(10)\, p}, \qquad 10^{-4} \le p < 0.1. \tag{B.81}$$

Equal logarithmic intervals receive equal weight; this is a documented project choice, not a supplied external rule.